UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

MARINUS PHARMACEUTICALS, INC.,
Petitioner,

v.

OVID THERAPEUTICS, INC.,
Patent Owner.

PGR2023-00020
Patent 11,395,817 B2

Before JEFFREY N. FREDMAN, ZHENYU YANG, and TIMOTHY G. MAJORS, *Administrative Patent Judges.*

FREDMAN, *Administrative Patent Judge.*

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 328(a)

I. INTRODUCTION

A. Background and Summary

Marinus Pharmaceuticals, Inc. ("Petitioner") filed a Petition (Paper 2, "Pet.") requesting post-grant review of claims 1–31 (the "challenged claims") of U.S. Patent No. 11,395,817 B2 (Ex. 1001, "the '817 patent"). Ovid Therapeutics, Inc. ("Patent Owner") filed a Preliminary Response. Paper 6 ("Prelim. Resp."). Patent Owner "disclaimed claims 1-21 and 23-24 of the '817 patent under 35 U.S.C. § 253(a) in compliance with 37 C.F.R. § 1.321(a)." Prelim. Resp. 4 (citing Ex. 2001, 1).

We determined, based on the information presented in the Petition and Preliminary Response, that the '817 patent was eligible for post-grant review, and that at least some of the challenged claims were more likely than not unpatentable at least based on Petitioner's challenges under 35 U.S.C. §§ 102, 103, and 112, first paragraph, as anticipated, obvious, and/or lacking enablement. Paper 9 ("Institution Dec."). Pursuant to 35 U.S.C. § 324, the Board instituted trial on August 17, 2023, of all challenged claims (claims 22 and 25–31). *Id*.

Following institution, Patent Owner filed a Response to the Petition (Paper 21, "PO Resp."), Petitioner filed a Reply to Patent Owner's Response (Paper 25, "Reply"), and Patent Owner filed a Sur-reply (Paper 29, "Sur-Reply"). Both Petitioner and Patent Owner filed various Objections to evidence (Papers 11, 23, 26) as well as Motions to Exclude certain evidence. Papers 34 and 35. An oral hearing was held on May 22, 2024, and a transcript has been entered into the record (Paper 40, "Tr.").

We have jurisdiction under 35 U.S.C. § 6(b). This Final Written Decision is issued pursuant to 35 U.S.C. § 328(a). Based on the record before us, we conclude that Petitioner has demonstrated by a preponderance of the evidence that claims 22 and 25–31 of the ’817 patent are unpatentable.

B. Real Parties-in-Interest

Petitioner identifies Marinus Pharmaceuticals, Inc. as the real party-in-interest. Pet. 91. Patent Owner identifies Ovid Therapeutics, Inc. as the real party-in-interest. Paper 4, 1.

C. Related Proceedings

Petitioner states, regarding related matters, that they are "not aware of any disclaimers, reexamination certificates, other than the PGR petition." Pet. 91. Patent Owner states that "U.S. Patent No. 11,395,817 (the ’817 patent) is not involved in any related litigation matters." Paper 4, 1.

II. THE ’817 PATENT

A. Background

The ’817 patent issued July 26, 2022, from US application 17/024,127 filed Sept. 17, 2020. Ex. 1001, codes (45), (21), (22). The ’817 patent is a continuation of US Application 16/789,709, filed Feb. 13, 2020, which is a continuation of US 16/447,300 filed June 20, 2019, which is a continuation of US 16/356,517 filed Mar. 18, 2019, which is a continuation of US 16/013,500 filed June 20, 2018, which is a continuation of US 15/673,737, filed Aug. 10, 2017, which claims the priority benefit of provisional applications US 62/490,293 filed Apr. 26, 2017, and US 62/373,589 filed Aug. 11, 2016. *Id.* at code (63), (60).

The ’817 patent "provides pharmaceutical compositions and methods that may be used in applications of epileptic disorders, such as status epilepticus." Ex. 1001, 2:7–9. The ’817 patent teaches "[s]tatus epilepticus is a condition in which seizures follow one another without recovery of consciousness between them. Accordingly, in embodiments, the disclosed methods are used to treat subjects that would otherwise be resistant to one or more conventional therapies." *Id.* at 10:8–42.

The '817 patent teaches "[i]n embodiments, methods are provided for treating an epileptic disorder by administering ganaxolone to a patient in need thereof." Ex. 1001, 6:27–29. The '817 patent teaches "methods of treating an epileptic disorder wherein the patient is provided improvement of at least one symptom for more than 4 hours after administration of the pharmaceutical composition to the patient." *Id.* at 14:21–25. The '817 patent further explains that "[i]n embodiments, improvement of at least one symptom for more than, e.g., 8 hours, 10 hours, 12 hours, 15 hours, 18 hours, 20 hours, or 24 hours after administration of the pharmaceutical composition to the patient is provided." *Id.* at 14:28–32.

B. Challenged Claims

Because Patent Owner disclaimed some claims, claims 22 and 25–31 are the challenged claims in the '817 patent, and are reproduced below. Because the claims all depend from claim 21 (now disclaimed), we also include claim 21 below for necessary context.

21. A method of treating status epilepticus comprising administering to a patient in need thereof a pharmaceutical composition comprising ganaxolone or a pharmaceutically acceptable salt thereof wherein the pharmaceutical composition is administered intravenously.

22. The method of claim 21 wherein the pharmaceutical composition provides improvement to the patient after administration for more than 8 hours.

25. The method of claim 21 wherein the patient is administered about 500 mg to 1000 mg/day ganaxolone.

26. The method of claim 21 wherein the patient is administered about 625 mg to 650 mg/day ganaxolone.
27. The method of claim 21 wherein the patient is administered about 650 mg to 675 mg/day ganaxolone.
28. The method of claim 21 wherein the patient is administered about 675 mg to 700 mg/day ganaxolone.
29. The method of claim 21 wherein the patient is administered about 625 mg to 700 mg/day ganaxolone.
30. The method of claim 21 wherein the patient is administered about 700 mg to 725 mg/day ganaxolone.
31. The method of claim 21 wherein the patient is administered about 1025 mg to 1050 mg ganaxolone.

Ex. 1001, 40:9–33.

III. INSTITUTED GROUNDS

Because Patent Owner disclaimed some claims, mooting some of Petitioner's grounds, we instituted trial based on all of the remaining applicable challenges to the patentability of the remaining claims of the '817 patent as presented in the Petition. *See, e.g.*, Pet. 4. These grounds are presented below.

Ground	Reference(s)/Basis	35 U.S.C. §	Claim(s) Challenged
1	Zhang[1]	§ 102	22, 25–31
2	Zhang	§ 103	22, 25–31
3	Zhang, Saporito P1[2]	§ 103	22, 25–31

[1] Zhang et al., WO 2016/127170 A1, published Aug. 11, 2016. Ex. 1009.
[2] Saporito et al., *Ganaxolone Administered Intravenously Prevents Behavioral Seizures and Promotes Survival in the Rat Lithium-Pilocarpine Model of Status Epilepticus*, Poster presented April 17, 2016, at the 2016 AAN Annual Meeting ("Saporito P1"). Ex. 1013.

	Saporito P2[3]		
4	Zhang, Marinus Orphan Drug Press Release,[4] Marinus Phase I Press Release[5]	§ 103	22, 25–31
7	Enablement	§ 112	22, 25–31

Petitioner relies on Declarations by Michael A. Rogawski, M.D. (Exs. 1003, 1041), Michael Saporito, Ph.D. (Exs. 1015, 1033), and Sharon Baughman, Ph.D. (Exs. 1058, 1060). Patent Owner relies on a Declaration by Raman Sankar, M.D., Ph.D. (Ex. 2014).

Based on the statements of qualifications and curricula vitae, we find Dr. Rogawski, Dr. Saporito, Dr. Baughman, and Dr. Sankar qualified to provide technical opinions from the perspective of a person of ordinary skill in the art ("POSA") in this proceeding. *See* Ex. 1003 ¶¶ 3–15; Ex. 1015 ¶¶ 2, 3, 8, 9; Ex. 1058 ¶¶ 6–13; Ex. 2014 ¶¶ 2–10. We address the admissibility of the Saporito and Baughman Declarations in our Order on Patent Owner's Motion to Exclude, which is entered concurrently with this Decision.

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[3] Saporito et al., *Intravenous Administration of Ganaxolone Attenuates Electroencephalographic Seizures in a Diazepam Resistant Model of Status Epilepticus*, Poster presented April 19 and April 20, 2016, at the 2016 AAN Annual Meeting ("Saporito P2"). Ex. 1014.
[4] *Marinus Pharmaceuticals Receives FDA Orphan Drug Designation for Ganaxolone IV to Treat Status Epilepticus*, Published April 15, 2016, Available at: https://ir.marinuspharma.com/news/news-details/2016/Marinus-Pharmaceuticals-Receives-FDA-Orphan-Drug-Designation-for-Ganaxolone-IV-to-Treat-Status-Epilepticus/default.aspx ("Marinus Orphan Drug Press Release"). Ex. 1011.
[5] *Marinus Pharmaceuticals Doses First Subject in Phase I Clinical Trial for Ganaxolone IV*, Published on June 22, 2016, Available at: https://ir.marinuspharma.com/news/news-details/2016/Marinus-Pharmaceuticals-Doses-First-Subject-in-Phase-1-Clinical-Trial-for-Ganaxolone-IV/default.aspx ("Marinus Phase I Press Release"). Ex. 1022.

IV. LEVEL OF ORDINARY SKILL IN THE ART

We consider the grounds of unpatentability in view of the understanding of a person of ordinary skill in the art as of the effective filing date of the challenged claims. Petitioner contends that one of ordinary skill in the art would

> have had a medical degree, or a Ph.D. in pharmacology or other field related to the development of pharmaceuticals, or equivalent degree, and several years of experience as a practicing neurologist treating patients who suffer from seizure disorders including status epilepticus and/or several years of experience in researching treatments for such seizure disorders.

Pet. 33–34. Patent Owner does not dispute Petitioner's definition of a skilled artisan, and provides no definition of its own in the Patent Owner Response or Patent Owner Sur-Reply. *See* Papers 21, 29.

Because Petitioner's proposed level of ordinary skill in the art is undisputed and consistent with the cited prior art, we adopt it for purposes of this Decision.

V. CLAIM CONSTRUCTION

In a *post-grant* review, we interpret a claim "using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b)." 37 C.F.R. § 42.200(b). Under this standard, we construe the claim "in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent." *Id.*

A. Interpretation of "Patient"

1. Petitioner's position

Petitioner "submits a construction for the term 'patient.' All remaining terms should be given their plain meaning." Pet. 34. Petitioner asserts

> The ’817 defines “patient” (used interchangeably with “subject”) as “includ[ing], but not limited to, primates such as humans, canines, porcine, ungulates, rodents, poultry, and avian.” One such patient within the scope of the Challenged Claims is, therefore, a rat. Another is a human.

Id. at 34–35 (citing Ex. 1001, 32:65–67; Ex. 1003 ¶ 67) (footnotes omitted, alteration in original). Petitioner asserts that if Patent Owner “wanted to prosecute claims limited to ‘treating a human patient,’ they could have done so (and are doing so in other applications in this family). But here, they didn’t.” Reply 6–7.

Petitioner asserts the “veterinary market was and continues to be significant for SE [status epilepticus], and the inclusion of various animal species in the definition of ‘patient’ is consistent with the understanding of the term in the art.” Reply 5. Petitioner asserts “[n]one of the limitations of claims 22 or 25-31 suggest any limitation on the term ‘patient’ either. Any patient may exhibit ‘improvement’ as recited in claim 22.” *Id*. at 6.

2. *Patent Owner’s position*

Patent Owner asserts “‘[t]reating status epilepticus comprising administering to a patient in need thereof’ in claims 22 and 25-31 means ‘treating a human patient suffering from SE.’ This construction is supported by the intrinsic record.” PO Resp. 17 (citation omitted); *cf.* Sur-Reply 4. Patent Owner contends that when claim 22 recites “a patient in need thereof,” a skilled artisan would thus understand that the “‘someone “in need”’ would be a human patient suffering from SE, and is not a rat or other animal receiving ganaxolone for research purposes.” PO Resp. 18.

Patent Owner asserts the Specification “refers to ‘treating’ in the context of addressing *clinical* and *subclinical* symptoms of a disease or a condition, and not in the context of preclinical, research uses." PO Resp. 18 (citing Ex. 1001, 29:29–51; Ex. 2014 ¶ 59). Patent Owner asserts that the "specification does not correlate 'treating' and 'treatment' to using rats for research purposes." *Id*. (citing Ex. 2014 ¶ 59). Patent Owner contends that the Examples "are simply 'administered' to rats. These examples do not refer to treating the rats." *Id*. at 19.

Patent Owner further contends that "for claims 25-31, a skilled artisan would understand that the doses recited in these claims (ranging from 500 mg to 1050 mg) are at least 130 times larger than the doses administered to rats in the '817 patent's examples." PO Resp. 19. Patent Owner asserts that "administering a 500 mg dose of ganaxolone to a rat would not fall within the range disclosed in the specification. A POSA thus would have understood that the 'patient' being administered the dosage amounts recited in claims 25-31 would not have been a rat." *Id*. at 20–21 (citing Ex. 2014 ¶ 63). Patent Owner asserts that a "POSA also would have understood that administering several thousand mg/kg ganaxolone to a rat would not result in treating SE and would instead be toxic to the rat." *Id*. at 22 (citing Ex. 2014 ¶ 64). Patent Owner concludes that "interpreting 'patient' in claims 25-31 to include a rat would lead to a 'nonsensical reading' of claims 25-31." *Id*.

Patent Owner contends

> Petitioner focuses on a single passage in the '817 patent that simply states that "[p]atient ... *include[s]*" multiple species. Ex. 1001, 32:65-67 (emphasis added). "Include" does not mean "require." And, as shown above, "patient" must be read in the context of its use, which in the challenged claims refers to treating a human patient in need of treatment for SE. In the

> context of claims 22 and 25-31, the recited doses clearly meant for administration to humans.

PO Resp. 23.

3. Analysis

We begin with the language of the claims. All of the claims recite the term "patient" and claim 21 specifically states "patient in need thereof." None of the claims at issue place any limitations on who can be a "patient."

We next turn to the Specification because "the specification 'is always highly relevant to the claim construction analysis. Usually, it is dispositive; it is the single best guide to the meaning of a disputed term.'" *Phillips v. AWH Corp.*, 415 F.3d 1303, 1315 (Fed. Cir. 2005) (en banc) (citing *Vitronics Corp. v. Conceptronic, Inc.*, 90 F.3d 1576, 1582 (Fed. Cir. 1996)). *Phillips* explains that "the specification may reveal a special definition given to a claim term by the patentee that differs from the meaning it would otherwise possess. In such cases, the inventor's lexicography governs." *Id.* at 1316.

> [T]he definition in the patent documents controls the claim interpretation. . . . Any other rule would be unfair to competitors who must be able to rely on the patent documents themselves, without consideration of expert opinion that then does not even exist, in ascertaining the scope of a patentee's right to exclude.

Southwall Tech., Inc. v. Cardinal IG Co., 54 F.3d 1570, 1578 (Fed. Cir. 1995).

The Specification of the '817 patent has a special definition where the inventor provided their own lexicography, and the definition states: "'Patient' and 'subject' are used interchangeably herein and include, but not limited to, primates such as humans, canines, porcine, ungulates, rodents, poultry, and avian." Ex. 1001, 32:65–67.

As we review the extrinsic evidence, a preponderance of the extrinsic evidence does not support limiting the term "patient" to human. While Dr. Sankar states that the ordinary artisan would have understood patient to mean "treating a human patient suffering from SE [status epilepticus]," Dr. Sankar never confronts in his Declaration the express definition of patient recited in the '817 patent. Ex. 2014 ¶¶ 58–64. And while Dr. Sankar discusses doses given to rats, Dr. Sankar does not address veterinary treatments in other animals included in the express definition in the '817 patent.

In response to being asked whether he agreed that there was "no human data provided for treatment of SE with ganaxolone in the 817 patent," Dr. Sankar answered "yes." Ex. 1042, 42:17–21. Similarly, when asked "are you aware of nonhumans that are in need of treatment for SE," Dr. Sankar answered "[t]here may be in veterinarian practice but that's outside my scope. But in my world we use the term 'patient' only for humans." *Id.* at 88:20–25. During deposition, Dr. Sankar was provided Exhibit 1051, titled "ACVIM[6] Consensus Statement on the management of status epilepticus and cluster seizures in dogs and cats" and acknowledged that these were nonhuman animals subject to diagnosis with SE. *Id.* at 89:18–25. When asked whether the doses in claims 25–31 could be administered to dogs or pigs, Dr. Sankar stated "depends on the size of the

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[6] ACVIM is the American College of Veterinary Internal Medicine. *See* Ex. 1051, 1.

dog" and "[a]gain, it depends on the size. I would say we're outside my realm of practice or even experiment." *Id*. at 92:4–8.

Dr. Rogawski states, in response to the definition of "patient" in the '817 patent that "[t]his definition indicates that a patient may be a human patient, or may be a non-human animal patient, including, for example, a dog (canine), pig (porcine), or rat (rodent)." Ex. 1041 ¶ 22. Dr. Rogawski states a "veterinarian's patient is just as much a patient as a medical doctor's patient, and veterinarians routinely treat their animal patients for SE using medications." *Id*. (citing Exs. 1050, 1051).

Dr. Rogawski notes that the distinction Patent Owner is asserting between "treating" humans and "administering" to rats collapses even within the context of the '817 patent itself, where

> the term "treat" is used specifically in relation to administering ganaxolone to a rat suffering from SE for the purpose of treating the SE (e.g., "dose/treatment/group" captions to Fig. 2 and Fig. 7; "Post-Treatment Seizures" title of Fig. 5; "Treatment" axis label of Fig. 6A; "Treatment", caption to Fig. 6B; column 35, lines 62–3, and 67; column 36, lines 14 and 42; column 37, lines 8, 40, and 65; and column 38, lines 13, 18, and 33).

Ex. 1041 ¶ 27. Dr. Rogawski states that "dogs are susceptible to status epilepticus and when this occurs, they are in need of treatment. Many types of dogs have body weights that would be similar to the weight range of humans" and "[t]hat the specific doses set forth in claims 25–31 may be higher than the dose that would likely be given to a rat does not suggest to a person of skill in the art that the term 'patient' in the claims is limited to humans." *Id*. ¶ 30. In deposition, Dr. Rogawski acknowledges that he is not an expert in veterinary science, but states "I treat animals very frequently in the laboratory setting. Much of my work involves studies on status epilepticus in animals. Also, I have collaborators with veterinarians, particularly at the University of Minnesota, who treat animals routinely with status epilepticus, and I confer with them on a regular basis." Ex. 2024, 26:23–27:16 (emphasis omitted).

As to Patent Owner's reliance on the terms "clinical" or "subclinical" to infer treatment only of humans, we find no persuasive reason or evidence explaining why treatment of animals is excluded by these terms. Patent Owner does not identify any discussion of any understanding the Examiner inferred from the prosecution history or any indication that the prosecution history disagrees with the express definition of "patient" in the Specification. The terms "clinical" and "subclinical" do not appear in claims 21, 22, or 25–31 nor are these terms defined by the '817 patent in any way that suggests the terms exclude non-human animals. *See, e.g.*, Ex. 1001, 29:29–51.

While a term in a Specification may be interpreted by implication, we are not persuaded by Patent Owner's attempt to infer a different construction that contradicts the express definition of "patient" and "subject" in the '817 patent. The extrinsic evidence provides no basis to exclude veterinary uses of the compound, particularly when the '817 patent expressly defines the term "patient" to include such veterinary patients.

As to the dosing amounts recited by claims 25–31, we credit Dr. Rogawski's statements that these doses would reasonably be expected to fall within the range required for certain veterinary patients such as dogs and cats of appropriate size, particularly as the weight range of humans from babies to adults overlaps the weight ranges of dogs and cats. *See, e.g.*, Ex. 1041 ¶ 30.

We therefore conclude that a preponderance of the evidence of record, including the definition recited in the ’817 patent and the extrinsic evidence, supports a finding that the terms “patient” and “subject” include animals such as rats, cats, and dogs, as well as humans.

B. Interpretation of “Improvement to the patient after administration for more than 8 hours”

1. Patent Owner’s position

Patent Owner asserts that:

> “Improvement to the patient after administration for more than 8 hours” in claim 22 means that there is improvement to the human patient more than 8 hours after the full dose of ganaxolone is given, regardless of whether the full amount of ganaxolone is given in a single administration or multiple administrations. Ex. 2014, ¶ 65.

PO Resp. 24. Patent Owner asserts “the claim requires measuring improvement *after* that total amount has been administered.” *Id*. Patent Owner cites Dr. Sankar as explaining “a POSA would have understood that whether the administration of a particular drug results in sustained improvement to a patient with SE would be assessed after the full dose of the drug is administered to the patient.” *Id*. (citing Ex. 2014 ¶ 66); *cf.* Sur-Reply 6.

2. Petitioner’s position

Petitioner asserts that claim 22 “does not require any specific dosing regimen.” Reply 7. Petitioner asserts

> The ’817 patent explains that “improvement” refers to a relative change in symptoms, which may be measured “in a subject prior to treatment, and again one or more times after treatment is *initiated*.” *Initiated*, not completed. This is consistent with how those in clinical practice measure improvement during treatment.

Id. (citing Ex. 1041 ¶¶ 32–37) (footnote omitted). Petitioner asserts the phrase "Improvement to the patient after administration for more than 8 hours" in claim 22 "simply requires measuring symptoms at a time more than 8 hours after treatment is initiated, as the '817 describes." *Id*. at 8. Petitioner asserts "*Initiated*, not completed. This is consistent with how those in clinical practice measure improvement during treatment." *Id*. at 7.

3. Analysis

We start with the language of claim 22 that recites "the pharmaceutical composition provides improvement to the patient after administration for more than 8 hours." Claim 22 does not indicate whether this is after the start of administration, after the completion of administration, or after some intermediate point. However, the plain language suggests that recited "improvement" occurs "after" the recited "administration" period. *See, e.g.*, *KEYnetik, Inc. v. Samsung Electronics Co., Ltd.*, 837 F. App'x 786, 792 (Fed. Cir. 2020) ("use of the word 'after' in the claim indicates a temporal relationship.")

We therefore turn to the '817 patent specification for clarification. The '817 patent repeats the claim language in a number of locations. *See, e.g.*, Ex. 1001, 5:1–2, 5:9–11; 14:21–41; 15:18–19; 24:45–49. The '817 patent provides definitions of "Improvement" and of "Improvement in next functioning," but neither of these definitions addresses the timing issue. *See id*. at 29:57–30:5. The schematic drawing in Figure 2, however, addresses timing, and is reproduced below:

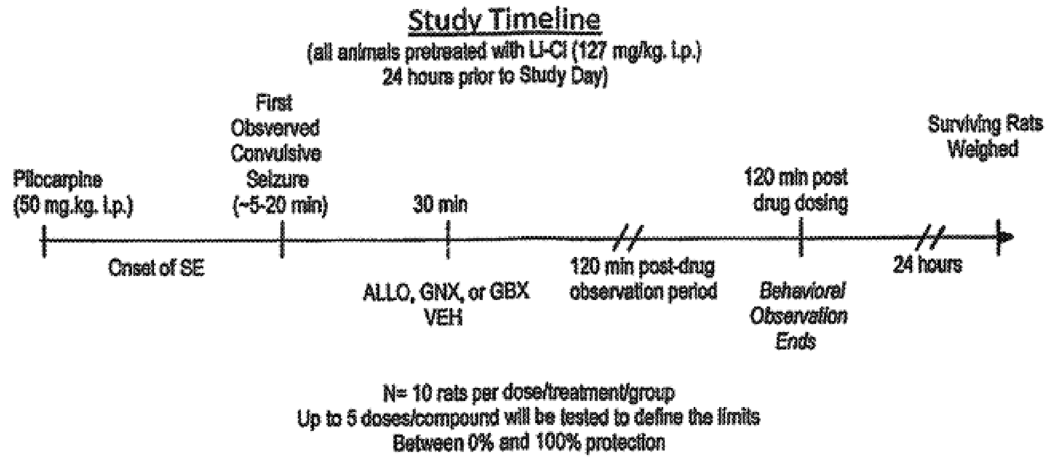

FIG. 2

"FIG. 2 is a schematic drawing depicting a timeline for an evaluation study of the ability of allopregnanolone, ganaxolone and gaboxadol to block benzodiazepine resistant status epilepticus in rats." Ex. 1001, 3:48–51.

Figure 2 differentiates between a 30 minute administration period, a 120 minute post-drug dosing period during which the animals are observed, and a 24 hour timepoint after which the animals are weighed, consistent with the '817 patent disclosure that "[a]ll rats are observed and scored for seizure severity for 120 min post drug administration." Ex. 1001, 35:64–65. This disclosure tends, on balance, to support Patent Owner's proposed interpretation.

Turning to the extrinsic evidence, Dr. Sankar similarly stated

> whether a patient with SE has sustained any improvement after administration of a drug is typically assessed at a point in time *after a full dose is administered*, rather than starting the clock after the first administration of only part of the dose. This is because the full dose of the drug is generally required to provide the expected effect.

Ex. 2014 ¶ 66 (emphasis added).

Dr. Rogawski, in contrast, stated

> measuring an improvement at intervals after initiation of treatment is consistent with how improvement is measured in clinical practice when SE is treated. Patients exhibiting SE are critically ill and they are monitored closely and continuously for changes in their condition after the start of treatment, not only after a treatment ends.

Ex. 1041 ¶ 34. In deposition, when asked "[w]ould you agree that Claim 22 requires improvement to the patient for more than eight hours after the total amount required to treat SE is administered?" Dr. Rogawski answered "I do not have an opinion on that." Ex. 2015, 46:9–13.

As we balance the plain language of claim 22, the intrinsic evidence in Figure 2 of the '817 patent, and the expert declarations, we find that a preponderance of the evidence better supports a conclusion that the skilled artisan would understand "improvement to the patient after administration for more than 8 hours" as improvement measured starting 8 hours after the complete dose of the compound is administered.

V. ANALYSIS

A. Principles of Law

In a post-grant review, as in an inter partes review, "the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable." *See Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016). This burden of persuasion never shifts to Patent Owner. *See Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015).

"Determining whether claims are anticipated involves a two-step analysis. The first step involves construction of the claims of the patent at issue." *In re Aoyama*, 656 F.3d 1293, 1296 (Fed. Cir. 2011). "The second step [of an anticipation analysis] involves comparing the claims to the prior art. Anticipation is a question of fact reviewed for substantial evidence." *Id.* "A prior art reference anticipates a patent claim . . . if it discloses every claim limitation." *In re Montgomery*, 677 F.3d 1375, 1379 (Fed. Cir. 2012). A reference may anticipate inherently if a claim limitation that is not expressly disclosed "is necessarily present, or inherent, in the single anticipating reference." *Verizon Servs. Corp. v. Cox Fibernet Va., Inc.*, 602 F.3d 1325, 1337 (Fed. Cir. 2010).

The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) objective evidence of nonobviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966). The obviousness inquiry also typically requires an analysis of "whether there was an apparent reason to combine the known elements in the fashion claimed by the patent at issue." *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 418 (2007) (citing *In re Kahn*, 441 F.3d 977, 988 (Fed. Cir. 2006) (requiring "articulated reasoning with some rational underpinning to support the legal conclusion of obviousness")). A petitioner cannot prove obviousness with "mere conclusory statements." *In re Magnum Oil Tools Int'l, Ltd.*, 829 F.3d 1364, 1380 (Fed. Cir. 2016). Rather, a petitioner must articulate a sufficient reason why a person of ordinary skill in the art would have combined the prior art references. *In re NuVasive*, 842 F.3d 1376, 1382 (Fed. Cir. 2016).

We analyze the asserted grounds of unpatentability in accordance with these principles.

B. *Overview of the Asserted Prior Art*

1. *Zhang (Ex. 1009)*[7]

Zhang is a published patent application filed with the World Intellectual Property Organization on Feb. 8, 2016 and published on August 11, 2016. Ex. 1009, codes (22), (43). Zhang discloses that "[s]tatus epilepticus (SE) is a serious seizure disorder in which the epileptic patient experiences a seizure lasting more than five minutes, or more than one seizure in a five minute period without recovering between seizures." *Id.* ¶ 1. Zhang teaches "[t]here are currently no good treatment options for these patients. The mortality rate for refractory status epilepticus (RSE) patients is high." *Id.* Zhang teaches "there exists the need for additional treatments for seizure disorders such as status epilepticus . . . . This disclosure fulfills this need and provides additional advantages." *Id.* ¶ 4.

Zhang teaches "an injectable ganaxolone formulation comprising ganaxolone, sulfobutyl ether-β-cyclodextrin; and water. . . . The disclosure also provides a method of treating a patient having a seizure disorder . . . comprising administering an effective amount of the injectable ganaxolone formulation." Ex. 1009 ¶ 6. Zhang teaches a "'patient' is a human or non-human animal in need of medical treatment. Medical treatment includes treatment of an existing condition, such as a disorder." *Id.* ¶ 22. "Seizure disorders that may be treated with the substituted β-cyclodextrin-ganaxolone injectable formulation include status epilepticus." *Id.* ¶ 70. Zhang teaches embodiments where the

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[7] The pagination used herein follows that in Ex. 1009, not the original pagination of the document but the paragraph numbers are those of the original document.

> substituted β-cyclodextrin-ganaxolone injectable formulation is administered as an intravenous infusion dose, either without [*sic*] or without a previous bolus dose for 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 consecutive days. The infusion dose may be administered at a rate of 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 mg/kg/hr or in a range of about 1 mg/kg/hr to about 10 mg/kg/ hr or 2 mg/kg/hr to about 8 mg/kg/hrs.

Id. ¶ 76.

Zhang provides a working example addressing ganaxolone effect on pilocarpine-induced status epilepticus in rats in Example 8. Ex. 1009 ¶ 117. Zhang teaches a study to "determine the ability of a CAPTISOL-ganaxolone (GNX) formulation administered intravenously (IV) at 12 and 15 mg/kg to inhibit pilocarpine induced status epilepticus (SE). . . . In this study, test drugs were administered either 15 or 60 minutes after onset of SE and compared to vehicle (30% Captisol)." *Id.* Zhang teaches "[c]ortical EEG[8] activity was recorded in rats before and after administration of pilocarpine to induce SE. Animals were pre-treated with LiCl to enhance and improve reliability of seizure induction and scopolamine to reduce peripheral cholinergic side effects of pilocarpine." *Id.* ¶ 118. Zhang teaches "EEG activity was recorded for 5 h after SE onset, and anticonvulsant activity was assessed by determining changes in EEG power over frequency ranges from 0.3 to 96 Hz." *Id.* Zhang teaches

> [t]he time of SE onset was determined by observation and recorded, and the animal dosed with the vehicle, allopregnanolone, or ganaxolone either 15 or 60 min post SE-Onset. Rats were observed for behavior changes at approximately 2, 15, 30, 60, 120 and 240 minutes after

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[8] "[B]rain waves are measured with an electroencephalogram [EEG]." Ex. 1001, 11:9–10.

> pilocarpine administration. Each animal was recorded once and euthanized 5 h after treatment.

Id. ¶ 127. Zhang teaches "[g]anaxolone at 12 and 15 mg/kg produced qualitatively similar results, with both dose levels producing a very strong reduction in SE amplitude overall." *Id.* ¶ 134. Table 6 of Zhang, comparing treatment with the control (vehicle) and 15 mg/ml ganaxolone administered 15 minutes after status epilepticus onset is reproduced in part below.

TABLE 6 - Treatment Effects, Rats Treated 15 minutes post SE onset							
		Significance, Time post drug administration					
Comparison	Frequency	0-15 min.	15-30	30-60	1-2 hrs	2-3 hrs	3-5 hrs
Vehicle vs GNX (12 mg/ml) both administered 15 minutes post SE onset	0-5 Hz	****	****	****	****	***	**
	5-10 Hz	****	****	***	**	*	ns
	10-30 Hz	***	****	**	*	ns	ns
	30-50 Hz	***	***	**	*	ns	ns
	50-70 Hz	**	**	*	**	ns	ns
	70-96 Hz	**	**	*	**	ns	ns
Vehicle vs GNX (15 mg/ml) both administered 15 minutes post SE onset	0-5 Hz	****	****	****	****	****	***
	5-10 Hz	****	****	****	****	***	ns
	10-30 Hz	****	****	****	**	*	ns
	30-50 Hz	***	***	***	**	ns	ns
	50-70 Hz	**	**	**	**	ns	ns
	70-96 Hz	**	***	**	**	ns	ns

"TABLE 6 summarizes the treatment effects for allopregnanolone or ganaxolone administered 15 minutes post seizure onset." Ex. 1009 ¶ 131. Table 6 shows that

> [w]hen dosed 15 minutes after SE-onset, GNX reduced EEG power across all frequency ranges to levels at or below baseline for at up to 5 h. From 5–70 Hz, EEG power was reduced to near the baseline level and remained there for 5 h post dosing. At 0–5 Hz, EEG power was reduced below the baseline level for approximately 90 min, and at 70–96 Hz, EEG power remained below baseline for the entire 5 h.

Id. ¶ 135.

2. Saporito P1 (Ex. 1013)

Saporito P1 is a poster presented at the American Academy of Neurology 2016 Annual Meeting. Ex. 1012, 46. Saporito P1 teaches their "studies were conducted to evaluate the dose- and time-dependent therapeutic potential of intravenously, (IV) administered GNX for the treatment of CSE in this clinically translatable rodent model. Moreover, these studies measured behavioral seizures and survival, and complement studies that measured seizures by EEG activity." Ex. 1013, 1. Saporito P1 teaches "animals received GNX dose-levels of 6, 9, 12 mg/kg." *Id*. Saporito P1 teaches "[c]ompounds were administered via IV administration at time of CSE onset (time 0) and 15, 30, and 60 minutes after onset." *Id*. Saporito P1 teaches that "[a]fter compound administration, animals were observed for the presence or absence of convulsive seizures (stage 3 or greater) for an additional 2 hours. Animals that did not exhibit a convulsive seizure during that time period were considered protected." *Id*. Saporito P1 concluded that

> Lithium-pilocarpine administration produced a convulsive status epilepticus. . . . GNX IV halted CSE and produced a dose-dependent reduction in seizures associated with CSE when administered at 3 different doses over four separate time points after the first observed convulsive seizure. . . . Significant activity of GNX was observed at a dose of 6 mg/kg and maximal protection at dose levels between 9 and 12 mg/kg. . . . GNX IV promoted survival at rates similar to those observed with administration of allopregnanolone as measured at 24 hours after CSE onset in CSE induced rats.

Id.

3. Saporito P2 (Ex. 1014)

Saporito P2 is a poster presented at the American Academy of Neurology 2016 Annual Meeting. Ex. 1012, 99. Saporito P2 teaches "current studies were designed to compare the anti-epileptic activity of GNX and ALLO in this treatment resistant model of SE." Ex. 1014, 1. Saporito P2 teaches "SE was first identified by presence of large amplitude slow wave EEG activity and then behaviorally using the Racine scale. Racine stage 3 seizures were used as the time-marker for SE onset and subsequent treatment." *Id*. Saporito P2 teaches "GNX, ALLO, or diazepam were administered via intravenous (IV) bolus injection 15 or 60 minutes after detection of SE onset." *Id*. Saporito P2 teaches

> Ganaxolone (GNX) (12 and 15 mg/kg) or allopregnanolone (ALLO) (15 mg/kg) were administered 15 minutes after status epilepticus (SE) onset and EEG seizure response measured for 5 additional hours. GNX at both dose-levels produced a durable reduction in seizure response (≥ 5 hrs). ALLO produced a shorter duration response (≤ 1 hr).

Id. Saporito P2 concluded "GNX administration elicits a sustained (≥ 5 hrs) block of SE independent of treatment time after SE onset." *Id*.

4. *Marinus Orphan Drug Press Release (Ex. 1011)*

Marinus Orphan Drug teaches "the U.S. Food and Drug Administration (FDA) has granted Orphan Drug Designation to the intravenous (IV) formulation of its CNS-selective $GABA_A$ modulator, ganaxolone for the treatment of status epilepticus." Ex. 1011, 1.

5. *Marinus Phase I Press Release (Ex. 1022)*

Marinus Phase I Press Release

> announced that it has dosed the first subject in its Phase 1 clinical trial of ganaxolone IV, an intravenous (IV) formulation of Marinus' CNS-selective GABAA modulator, for the treatment of status epilepticus (SE). The study is designed to evaluate the safety, pharmacokinetics (PK) and

> Pharmacodynamics (PD) of ganaxolone IV in healthy volunteers.

Ex. 1022, 1.

C. Ground 1 – Anticipation over Zhang

Petitioner contends that claims 22 and 25–31 are anticipated by Zhang. Pet. 44–45, 46–48.

1. Petitioner's positions

a. Petitioner's position on claim 22

Petitioner asserts

> Zhang teaches a POSA how to make its ganaxolone formulations throughout the specification (including by way of working examples), and also teaches a POSA how to *use* its ganaxolone formulations to treat SE, in particular, by including working examples demonstrating the efficacy of ganaxolone for the treatment of SE in the rat lithium-pilocarpine model.

Pet. 36 (citing Ex. 1009 ¶¶ 116–136; Ex. 1003 ¶ 65) (footnote omitted). Petitioner asserts

> [e]xample 8 of Zhang describes the effects of ganaxolone in rats with lithium pilocarpine-induced SE.
>
> The rats of Example 8 were administered a CAPTISOL®-ganaxolone formulation intravenously at 12 and 15 mg/kg, administered either 15 or 60 minutes after onset of SE. Therapeutic activity of ganaxolone in the treatment of SE was assessed by determining changes in EEG power for five hours after onset of SE. Ganaxolone at both 12 and 15 mg/kg "produced a very strong reduction in SE amplitude overall."

Id. at 36–37 (citing Ex. 1009 ¶ 117, 118, 134; Ex. 1003 ¶ 66) (footnotes omitted). Petitioner specifically asserts, with regard to independent claim 21 from which claim 22 depends, "Zhang teaches that its ganaxolone formulations 'include parenteral formulations suitable for intravenous infusion,' and teaches a POSA how to use ganaxolone (intravenously) to treat a patient with SE (whether human or rat)." *Id.* at 44 (citing Ex. 1001, 38:48–51, 40:8–12; Ex. 1003 ¶¶ 89–90).

Petitioner asserts, specific to claim 22, that "[w]hile Zhang does not expressly disclose that its ganaxolone formulations provide 'improvement to the patient after administration for more than 8 hours,' this would be the natural result of administering them to a patient with SE, as taught by Zhang. As such, Zhang inherently teaches the limitation of claim 22." Pet. 45 (citing Ex. 1003 ¶ 93); *cf.* Reply 10.

Petitioner asserts, as a separate basis for Zhang anticipating claim 22, that

> Zhang states: "Methods of treatment also include administering multiple injections of the β-cyclodextrin-ganaxolone injectable formulation over a period of 1 to 10 days. The injections may be given at intervals of 1 to 24 hours." A skilled artisan would immediately recognize that if a single bolus treatment for SE suppresses seizures for up to 2 hours as ganaxolone is clearly shown to do in Tables 6 and 7, then the treatment could be repeated at intervals of 2 hours to provide sustained treatment for more than 8 hours.

Pet. 45 (footnote omitted).

b. Petitioner's position on claims 25–31

Petitioner asserts "Zhang discloses 'embodiments in which multiple bolus doses of the ganaxolone/sulfobutyl ether-β-cyclodextrin formulation are administered to the patient.'" Pet. 46–47 (citing Ex. 1009 ¶ 85; Ex. 1003 ¶ 102). Petitioner asserts that in "certain embodiments the multiple bolus doses are given over 1 to 10 days at intervals of 1 to 24 hours" and that "[i]n certain embodiments each bolus dose comprises about 1 mg/kg to about 20 mg/kg ganaxolone." *Id.* (alteration in original).

Petitioner's expert prepared a table correlating the dose ranges recited in claims 25–31 with the same dose range recited in amounts administered to a 70 kg patient in mg/kg and found that "Zhang's bolus doses of 1 mg/kg to 20 mg/kg, for a 70 kg patient, encompass each of the amounts recited in claims 24–31." Pet. 48 (citing Ex. 1003 ¶ 104). Petitioner asserts "[t]here is no 'picking and choosing' required to conclude that Zhang describes administering IV ganaxolone to a patient in the claimed amounts sufficiently to enable a POSA to do so." Reply 12.

2. *Patent Owner's positions*

a. *Patent Owner's position on claim 22*

Patent Owner asserts that Petitioner "points to nothing in Zhang that describes treating a human patient, let alone one suffering from SE." PO Resp. 25. Patent Owner asserts "[n]or can Petitioner point to any relevant disclosure in Zhang that describes intravenously administering ganaxolone to a human patient suffering from SE. . . . Thus, for this reason alone, Petitioner has not established that Zhang anticipates any of the challenged claims, all of which depend from claim 21." *Id.* at 26.

Patent Owner asserts "Petitioner points to no evidence as to why 'improvement to the patient after administration for more than 8 hours' would inevitably happen based on Zhang's disclosure." PO Resp. 30. Patent Owner asserts that "Zhang cannot demonstrate that ganaxolone has any effect on SE in a human patient, through inherency or otherwise, because ganaxolone was not administered to a human patient." *Id.* at 29. Patent Owner states a "showing of inherent anticipation requires that the limitation *inevitably* occur, not just might occur." *Id.* (citing *Allergan, Inc. v. Apotex Inc.*, 754 F.3d 952, 960–61 (Fed. Cir. 2014)). Patent Owner notes that

> Dr. Rogawski admitted that after the 5-hour mark in Zhang's Example 8 that "it's possible that there could be improvement, there might not be improvement. When one is treating a medical condition, there's always a degree of variability and one doesn't expect a certain outcome." Ex. 2015, 155:7-16; see also Ex. 2014, ¶¶ 76-78. This is insufficient to demonstrate anticipation by inherency.

Id. at 30.

Patent Owner asserts "Zhang instead discloses the opposite—any effect of ganaxolone in reducing EEG power *diminishes* by 5 hours." PO Resp. 31 (citing Ex. 1009 ¶¶ 135–136; Ex. 2014 ¶ 76). Patent Owner also asserts that "Petitioner's argument that the 8-hour limitation in claim 22 can be satisfied by repeated treatments of ganaxolone "at intervals of 2 hours" is also without merit" because the argument "contradicts the ordinary and customary reading of the claim language, which requires improvement to the patient for more than 8 hours after the *total dose* of ganaxolone required to treat SE has been administered." *Id.* at 32. Patent Owner further asserts "Petitioner improperly relies on what a POSA allegedly would have known to fill the gaps in Zhang," "Petitioner necessarily resorts to improper picking and choosing from different portions of Zhang to allegedly meet this limitation," and "Petitioner provides no evidence as to why improvement to the patient for more than 8 hours after the total dose of ganaxolone is administered would inevitably happen once the total dose is divided into multiple injections at 2-hour intervals." *Id.* at 33–35.

b. Patent Owner's position on claims 25–31

Patent Owner asserts

> [n]othing in Zhang ties its disclosure of bolus doses of "about 1 mg/kg to about 20 mg/kg" to any one disease state, let alone treating SE. Zhang's general disclosure is not specific to any one disease state, dosage strength, or dosing regimen. Ex. 2014 ¶ 82. Indeed, Zhang includes numerous options beyond dose amount—it also postulates that the ganaxolone formulations could be used to treat at least 31 different diseases or disorders.

PO Resp. 37. Patent Owner contends that Zhang "provides several broad dose ranges for administering a ganaxolone formulation in the abstract, and not for any specific disease. . . . This level of picking and choosing from disparate paragraphs in Zhang cannot support Petitioner's anticipation challenge." *Id*. at 38. Patent Owner asserts that Petitioner's "picking and choosing among Zhang's disclosure of 31 different diseases or disorders and various dosing regimens to arrive at the specific dose ranges for treating SE in the claims remains improper." Sur-Reply 11.

Patent Owner contends that Zhang's range allows for potential daily doses ranging from "as a weight-based dose range for a 70 kg adult human, about 70 mg/day to 33,600 mg/day. Ex. 2014, ¶ 83." PO Resp. 38. Patent Owner asserts that this "encompasses several broad dose ranges that would be ineffective to treat SE patients and not result in improvement to the patient as claims 25–31 require. Ex. 2014, ¶ 85." *Id*. Patent Owner notes that "Zhang's Example 8 provides data in only the rat-lithium pilocarpine model and only at ganaxolone doses of 12 mg/kg and 15 mg/kg. Ex. 1009, ¶ [0117]." *Id*.

3. Analysis

a. Claim 22

It is Petitioner's burden to show by a preponderance of the evidence that claim 22 of the '817 patent is unpatentable as anticipated by Zhang, and we find that Petitioner has failed to meet its burden. *See* 35 U.S.C. § 326(e).

We agree with Petitioner, as discussed in our claim interpretation section above, that patients within the scope of claim 22 include animals such as rats, and also that Zhang teaches "[s]eizure disorders that may be treated with the substituted β-cyclodextrin-ganaxolone injectable formulation include status epilepticus." Ex. 1009 ¶ 70. As to the intravenous administration requirement in claim 21, from which claim 22 depends, we also agree with Petitioner that Zhang teaches embodiments where the "substituted β-cyclodextrin-ganaxolone injectable formulation is administered as an intravenous infusion dose." *Id.* ¶ 76. We therefore agree that Zhang teaches intravenous treatment using a ganaxolone formulation in patients with status epilepticus. *See, e.g.*, *id.* ¶¶ 22, 70, 76.

As discussed above, we agree with Patent Owner that claim 22 is properly interpreted to require improvement 8 hours after total administration of the ganaxolone dose. A preponderance of the evidence of record does not support Petitioner's position that Zhang inherently, and therefore necessarily, provides improvement after administration for more than 8 hours. Even if claim 22 simply required improvement 8 hours after initiation of a ganaxolone dose, we would still conclude that a preponderance of evidence does not support Petitioner's position.

Zhang provides no evidence of improvement at an 8 hour timepoint because Zhang's working example in rats terminates at 5 hours as the rats are then sacrificed for analysis. Ex. 1009, 24 ¶ 127. Zhang states that in the experiment on rats with induced status epilepticus, ganaxolone "reduced EEG power across all frequency ranges to levels at or below baseline for at up to 5 [hours]." *Id.* ¶ 135. This is the only direct evidence in Zhang and suggests that ganaxolone continued providing improvement up to the 5 hour point, but provides no evidence regarding improvement after 5 hours.

As to whether Zhang's treatment of SE with ganaxolone would inherently provide improvement at an 8 hour timepoint, Petitioner's expert Dr. Rogawski stated "[t]o the extent that improvement to the patient were to occur after that time period, however, it would be the natural result of administering the ganaxolone formulations in the manner taught in Zhang." Ex. 1003 ¶ 93. However, in deposition, when asked "are you acknowledging that it's possible that the improvement may not be occurring after that time period?" Dr. Rogawski stated "I think it's possible that there could be improvement, there might not be improvement. When one is treating a medical condition, there's always a degree of variability and one doesn't expect a certain outcome." Ex. 2015, 155:9–16. Thus, Dr. Rogawski acknowledged that improvement at the 8 hour timepoint is not necessarily the natural result of ganaxolone treatment and may not necessarily occur.

We recognize the '817 patent's Specification shows that 10 and 20 mg/kg ganaxolone doses that bracket the 12 and 15 mg/kg ganaxolone doses tested by Zhang, result in improved 24-hour survival relative to lower doses. *See* Ex. 1001, Fig. 4; Ex. 1009 ¶ 117. And while this evidence in the '817 patent itself does tend to show that doses bracketing those recited in Zhang provide continued improvement to the rat patients for periods exceeding 8 hours, inherency "may not be established by probabilities or possibilities."

Par Pharm., Inc. v. TWI Pharm., Inc., 773 F.3d 1186, 1195 (Fed. Cir. 2014). Zhang alone does not demonstrate that doses of ganaxolone administered intravenously necessarily provide at least 8 hours of improvement to a patient or at a dose encompassed by claim 22.

As to the issue of whether Zhang's teaching to treat with multiple doses also teaches "improvement to the patient after administration for more than 8 hours" as required by claim 22, on this record, we agree with Petitioner that claim 22 does not distinguish between a single administration event or multiple administrations, and therefore is reasonably interpreted as broadly open to both single administration and multiple administrations. This understanding is consistent with the '817 patent, which teaches that "[g]anaxolone or a pharmaceutically acceptable salt thereof can be administered, e.g., once daily, twice daily, three times daily, or four times daily." Ex. 1001, 8:35–37.

We acknowledge that Zhang teaches "[d]osing schedules in which the substituted β-cyclodextrin-ganaxolone injectable formulation is injected every 1 hr, 2 hrs, 4 hrs, 6 hrs, 8 hrs, 12 hrs, or 24 hours are included herein." Ex. 1009 ¶ 74. However, Zhang does not provide examples or direct evidence showing that administration of any particular dose of ganaxolone every two hours necessarily results in 8 hour improvement in any particular species of patient.

Accordingly, we find that Petitioner does not show, by a preponderance of the evidence, that claim 22 of the '817 patent is unpatentable as anticipated by Zhang.

b. Claims 25–31

We find that Petitioner has failed to provide a preponderance of the evidence to show anticipation by Zhang of the ganaxolone dose ranges recited in claims 25–31.

The parties do not dispute that Zhang teaches a range of doses of about 1 mg/kg to about 20 mg/kg of ganaxolone that overlap with the recitation of ganaxolone ranges recited in claims 25–31. *See* Ex. 1009 ¶ 84; Ex. 1003 ¶ 103. The issue in the anticipation analysis is

> whether the prior art discloses a point within the claimed range or discloses its own range that overlaps with the claimed range. If the prior art discloses a point within the claimed range, the prior art anticipates the claim. *See Ineos USA LLC v. Berry Plastics Corp.*, 783 F.3d 865, 869 (Fed. Cir. 2015) . . . . On the other hand, if the prior art discloses an overlapping range, the prior art anticipates the claimed range "only [ ] if it describes the claimed range with sufficient specificity such that a reasonable fact finder could conclude that there is no reasonable difference in how the invention operates over the ranges." *Id.*

UCB, Inc. v. Actavis Labs. UT, Inc., 65 F.4th 679, 687 (Fed. Cir. 2023) (second alteration in original).

Dr. Rogawski's declaration addresses the anticipation issue in stating that "Zhang's bolus doses of 1 mg/kg to 20 mg/kg, for a 70 kg patient, encompass each of the amounts recited in claims 24–31. Therefore, in my opinion, Zhang anticipates claims 24–31." Ex. 1003 ¶¶ 104–105. Dr. Rogawaski was then asked in deposition about whether any specific points within the range were disclosed. Specifically, when asked about a calculation for mouse body weights that "the range that you get is below the 500-milligram dose that's the lowest dose recited in Claims 25 through 31 of the '817 Patent?," Dr. Rogawski stated that "based on your calculation, your

– I can – I can support your mathematical calculations." Ex. 2015, 161:16–23. That is, Dr. Rogawski acknowledged that the range for mouse body weights would not have included specific dose points within the ranges of claims 25–31 of the '817 patent. And then when further asked "Are you aware of any publications in which 500 milligrams per day or more of ganaxolone was administered to a rat to treat SE?" Dr. Rogawski stated "Off the top of my head, I'm not, no." *Id.* at 161:25–162:4. Thus, while Dr. Rogawski identifies an overlapping range, Dr. Rogawski acknowledged Zhang did not disclose a point within the ranges recited in claims 25–31 of the '817 patent.

Turning from whether there was a disclosure of points within the claimed range to whether the ranges of the prior art and the claims overlap, Dr. Sankar stated "Zhang's Example 8 provides data in the rat-lithium pilocarpine model and only at ganaxolone doses of 12 mg/kg and 15 mg/kg." Ex. 2014 ¶ 87. Dr. Sankar stated "doses of ganaxolone administered to rats cannot be directly compared with doses administered to humans because rats metabolize compounds at a much faster rate than humans by the hepatic microsomal system. As a result, higher doses (in mg/kg) of a compound are typically administered to rats than humans." *Id.*

Patent Owner cites an assessment of ganaxolone safety that states "Higher doses (40 mg/kg in mice and 20 mg/kg in rats) also caused a loss of righting reflex, which was considered adverse. Cognitive function was significantly impaired at a dose of 30 mg/kg. These GNX effects were presented as deficits in motor function or in the latency to learn a response and were attributed to the sedative effect of GNX." *See* PO Resp. 22 (citing Ex. 2017, 25). While the cited assessment is post-filing date, it does evidence that there is a reasonable difference in how the invention operates over the range in rats, where lower values may be effective and higher values may be toxic.

Patent Owner has therefore provided some evidence showing that picking and choosing particular doses from the ranges recited in Zhang would be required to provide a method of treatment. Because the doses recited in claims 25–31 of the ’817 patent would result in differences in how the doses of ganaxolone would function to treat status epilepticus patients, these different doses would consequently result in differences in the improvement or toxicity to the patient not disclosed in Zhang. Therefore the range recited in Zhang does not anticipate the significantly narrower ranges recited in claims 25–31 of the ’817 patent. *Cf. In re Arkley*, 455 F.2d 586, 587–588 (CCPA 1972) (An anticipatory reference "must clearly and unequivocally disclose the claimed compound or direct those skilled in the art to the compound without any need for picking, choosing, and combining various disclosures not directly related to each other.").

Accordingly, we find that Petitioner does not show, by a preponderance of the evidence, that claims 25–31 of the ’817 patent are unpatentable as anticipated by Zhang.

D. Ground 2 – Obviousness over Zhang

1. Petitioner positions

a. Petitioner's position on claim 22

Petitioner asserts

> If Zhang's teachings, as described in Section VIII.A, do not teach the limitations of claim 1–3, 10–17, and 19–31, they suggest them. And, based on Zhang's teachings, including, for example, the results described in Example 8, a POSA would

> have had a reasonable expectation of success at arriving at the claimed methods of treating SE with ganaxolone—whether in a rat patient or a human patient.

Pet. 49 (footnote omitted).

b. Petitioner's position on claims 25–31

Petitioner asserts

> Zhang describes treating SE by "administering an effective amount of [injectable ganaxolone] to a patient." Zhang teaches that in some embodiments the method of treatment is administration of a single bolus doses of from about 1-20 mg/kg, (which is 70-1,400 mg for a 70 kg human), as well as multiple bolus doses over 1 to 10 days at intervals of 1 to 24 hours in the same amount as the single bolus. Zhang's doses and, therefore, the doses of claims 25-31, are consistent with what a POSA already knew about the clinical pharmacology of ganaxolone.

Reply 14–15 (citing Ex. 1041 ¶ 54; Ex. 1009 ¶¶ 22, 69–89; Ex. 1003 ¶¶ 102–104) (footnote omitted, alteration in original).

c. Petitioner's position on objective indicia

Petitioner asserts:

> Because it did nothing to develop ganaxalone, Ovid [Patent Owner] tries to offer *Marinus's* [Petitioner's] clinical trial work as evidence of "unexpected results" of the Challenged Claims, all of which were added to Ovid's application after Marinus began to publish its clinical trial work. But Ovid conveniently ignores and/or misrepresents the details of Marinus's work, which are absent from Ovid's claims.

Reply 18 (citing Ex. 1041 ¶¶ 11–17; Ex. 1046; Ex. 1047) (footnote omitted). Petitioner asserts that the Patent Owner's theory of unexpected results were achieved with Petitioner's specific dosing regimens and "are nowhere near commensurate in scope with the Challenged Claims and, therefore, cannot be used as evidence of nonobviousness." *Id*. at 19.

Petitioner asserts that in the clinical trial, the

> three dosing cohorts were referred to as the "Low" "Medium," and "High" cohorts, all of which included a bolus for the "rapid loading of ganaxolone (plasma concentration ~ 900 ng/ml) designed to abort SE" followed by continuous infusion to administer "maintenance doses aimed at sustaining seizure control." After 2 days, if the patient was deriving continuing benefit, investigators were allowed to extend infusions for an additional 2 days (4 days total) before beginning the 18-hour taper.

Reply 21–22 (citing Ex. 1041 ¶ 65; Ex. 2003, 2383) (footnote omitted). Petitioner relied on this trial to show unexpected results for its own patent applications, and asserts that when the Hulihan[9] "declarations were submitted, the claims of both Marinus applications required that '*ganaxolone is administered as a bolus plus continuous infusion*' to maintain plasma concentrations of 500-1000 ng/mL for 8-12 hours. Ovid's claims contain no such limitations." *Id*. at 24 (citing Ex. 2007, 29–32; Ex. 2008, 25–28).

Petitioner asserts that "Claim 22 covers administration of ganaxolone in *any* amount, in *any* intravenous format, and in *any* patient. . . . Claim 22 does not require an initial bolus dose followed by a continuous infusion . . . Claim 22, with no limitation on dosing, is not commensurate in scope with Marinus's results." Reply 25–26. Petitioner also asserts "claims 25-31 do not require any particular dosing regimen, and certainly not the specific

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[9] Declarations of Dr. Joseph Hulihan, MD, dated July 7, 2021. At least at the time of the Declarations, Dr. Hulihan was the Chief Medical Officer for Marinus Pharmaceuticals. *See, e.g.*, Ex. 2007, 13.

regimens that Marinus used to achieve the results in its clinical trial." *Id.* at 28. Petitioner notes "Dr. Sankar acknowledged that the 500 ng/mL plasma concentration that led to the results is 'a function of the bolus and then the continuous infusion afterwards.'" *Id.* (citing Ex. 1042, 65:9–66:10).

Petitioner asserts as to long felt need that "it is Marinus [Petitioner], not Ovid [Patent Owner], who has worked for years to provide that treatment option, as evidenced by the cited art." Reply 30. Petitioner assets "[o]n failure of others, as the Board already stated, 'rather than show failure by others, the evidence of record suggests success by others.' And any supposed failure by non-ganaxolone drugs is irrelevant." *Id.* (citing Institution Dec. 35) (footnote omitted).

2. *Patent Owner's positions*

a. *Patent Owner's position on claim 22*

Patent Owner asserts "nowhere does Zhang describe 'improvement to the patient after administration for more than 8 hours.' *See also* Ex. 2014, ¶ 91. Neither the Petition nor Dr. Rogawski's declaration makes any attempt to demonstrate why a skilled artisan would have been motivated by Zhang to arrive at this limitation." PO Resp. 52. Patent Owner asserts "Petitioner's attempt to supply a motivation to fill in a limitation missing from Zhang through the 'common knowledge' of a POSA without any reasoned basis or evidentiary support is improper." *Id.* at 53.

Patent Owner asserts "Petitioner points to no evidence as to why a POSA would have had a reasonable expectation of success that intravenous administration of ganaxolone would result in 'improvement to the patient after administration for more than 8 hours' when Zhang itself shows an absence of any effect by 5 hours." PO Resp. 55. Patent Owner also asserts "a

POSA would not have had a reasonable expectation of success that ganaxolone treatment 'repeated at intervals of 2 hours' would result in 'improvement to the patient after administration for more than 8 hours.' Ex. 2014, ¶¶ 95, 98." *Id*.

b. Patent Owner's position on claims 25–31

Patent Owner asserts that a "POSA would not have been motivated to choose the specific dose ranges for treating SE recited in claims 25-31 from the broad dose range relied upon by Petitioner in paragraph 85 of Zhang." PO Resp. 57. Patent Owner asserts "Zhang provides no suggestion or motivation to choose the dose ranges recited in claims 25-31 from among the broad range of doses and treatment options described in Zhang. Ex. 2014, ¶¶ 101-102." *Id*. at 58.

Patent Owner asserts "Petitioner fails to show that a POSA would have had a reasonable expectation of success in treating SE with the dose ranges recited in claims 25-31 based on the disclosure of Zhang. Pet. 48-49; Ex. 2014, ¶¶ 103-106." PO Resp. 58. Patent Owner asserts

> A POSA would have understood that the effectiveness of a particular dose in the rat pilocarpine model does not directly translate to human treatment due to at least the differences in the onset of SE in rats (chemically induced) and humans (occurring by various clinical circumstances) and a rat's faster metabolism by the hepatic microsomal system. Ex. 2014, ¶ 71.

Id. at 59. Patent Owner asserts that Dr. Rogawski acknowledges "the rat pilocarpine model at most can predict that a compound potentially could be useful in the treatment of SE in humans. *See, e.g.*, Ex. 1003, ¶¶ 25" and that the "prediction often does not translate into demonstrated efficacy in humans, particularly in the neuroactive steroid class of compounds that ganaxolone belongs to. *See, e.g.*, Ex. 2015, 23:6-25:7." *Id*.

c. Patent Owner's position on objective indicia

Patent Owner asserts "[t]he methods of treating SE by administering ganaxolone recited in claims 22 and 25-31 were found to be unexpected and critical to the claimed invention. Ex. 2014, ¶ 107." PO Resp. 40. As evidence that these methods would have been unexpected, Patent Owner cites Declarations of Dr. Joseph Hulihan in US applications submitted by Petitioner that stated

> Marinus Pharmaceuticals Inc. is investigating the use of ganaxolone for treating SE. Given the complexities of SE and difficulties in treating SE, we initially believed that a *high ganaxolone exposure for a long treatment period of at least 4 days duration, would be required to break SE*, *i.e.*, stop seizures and prevent seizures from rapidly returning when ganaxolone was discontinued.

PO Resp. 42 (citing Ex. 2007, 30 ¶ 7 (emphasis added by Patent Owner); Ex. 2008 26 ¶ 7).

Patent Owner also points to a phase 2 clinical trial of Petitioner where "[f]or each dose cohort, 'ganaxolone infusion was initiated as an IV bolus (over 3 min) with continuous infusion of decreasing infusion rates for 48-96 h followed by an 18-h taper.' [Ex. 2003], 2381." PO Resp. 43. Patent Owner asserts that the study observed that

> "rapid" seizure cessation and reduction in seizure burden and lack of recurrence of SE after discontinuation of the ganaxolone infusion resulting from IV administration of 500 mg/day, 650 mg/day, and 713 mg/day were unexpected over Petitioner's and its scientist's (a person of ordinary skill) expectation that "high ganaxolone exposure for a long treatment period of at least 4 days duration, would be required to break SE."

Id. at 44 (citing Ex. 2007, 30 ¶ 7; Ex. 2008, 26 ¶ 7). Patent Owner asserts "it was unexpected that 82% of patients 'did not require treatment for SE with additional second-line IV ASMs or IV anesthetics through 24 h following completion of the ganaxolone infusion (Table 2)' and that . . . '85% of evaluable patients did not have SE relapse,' [Ex. 2003], 2386." *Id*. at 45.

Patent Owner asserts that it is "not required to show that every unrecited possibility provides the same unexpected results. [Patent Owner] must instead show only that the treatment or daily dose amounts recited in the claims provide unexpected results." Sur-Reply 13. Patent Owner asserts Petitioner "fails to show that something other than the treatment and daily dose amounts cause the unexpected results." *Id*. at 14. Patent Owner asserts

> Petitioner provided that only a dose of 713 mg/day allows for a plasma concentration ≥500 ng/mL for greater than 8 hours and sustained seizure reduction throughout the entire time period tested. As a dose of 713 mg/day falls squarely within the claimed range of 700 mg to 725 mg/day, these statements further confirm that the effects of the 713 mg/day dose on seizure burden reduction and cessation were unexpected over Petitioner's and its skilled artisan scientist's expectations and were markedly superior over the other doses encompassed by Zhang's extremely broad disclosure.

PO Resp. 48 (citing Ex. 2014 ¶¶ 113–116).

Patent Owner also asserts that the objective indicia of long-felt need and failure of others. Patent Owner asserts the "prior art is replete with statements recognizing the long-felt need for a drug that could treat SE. . . . Ganaxolone administered according to the methods in claims 22 and 25-31 filled this unmet need." PO Resp. 50. Patent Owner asserts their

> claimed methods stand against a stark backdrop of repeated failures for treating SE. Intravenous benzodiazepines, such as lorazepam and diazepam, are among the "first-line therapies" for treating SE, but as Petitioner admits, "[a] significant drawback to these therapies is that their efficacy decreases

> dramatically as the duration of SE increases, and there can be a complete loss of efficacy after a certain amount of time."

Id. at 51 (citing Ex. 1004, 18287) (citation omitted, alteration in original).

3. *Analysis*

We recognize Patent Owner asserts objective indicia in favor of patentability and Petitioner asserts the objective indicia do not support patentability. *See* PO Resp. 39–51; Reply 18–30.

a. *Claim 22*

We find that Petitioner has failed to provide a preponderance of the evidence to satisfy the burden to show that claim 22 of the '817 patent is unpatentable as obvious over Zhang alone.

Patent Owner correctly points out that Zhang lacks any express disclosure suggesting that treatment with ganaxolone will result in "improvement to the patient after administration for more than 8 hours." Also, as we discussed above with regard to anticipation, a preponderance of the evidence does not show that Zhang alone would inherently and necessarily result in 8 hour improvement in any particular species of patient after administration of ganaxolone. That conclusion is supported by Petitioner's own expert, Dr. Rogawski, who stated with regard to the claimed 8 hour improvement that "I think it's possible that there could be improvement, there might not be improvement. When one is treating a medical condition, there's always a degree of variability and one doesn't expect a certain outcome." Ex. 2015, 155:9–16. Dr. Rogawski therefore acknowledges that improvement at the 8 hour timepoint is not necessarily the natural result of ganaxolone treatment and may not necessarily occur.

Dr. Sankar, Patent Owner's expert, notes

> Zhang's Example 8 discloses that the effect of ganaxolone on reducing EEG power is largely diminished by 5 hours and that ganaxolone showed a significant reduction at 3-5 hours in only one low frequency group (0-5 Hz). Ex. 1009, ¶¶ [0135]-[0136]. Zhang provides no data regarding seizure reduction beyond 5 hours.

Ex. 2014 ¶ 94. Dr. Sankar concludes that "Zhang teaches nothing that would establish that ganaxolone provides seizure reduction for 8 hours after administration." *Id.* ¶ 97.

We appreciate that for obviousness, unlike anticipation, Zhang may render the claims obvious in the absence of inherency, so long as there would have been a reasonable expectation of success in obtaining improvement for a period of 8 hours after ganaxolone treatment. Dr. Rogawski states "even transient improvement (e.g., during the first five hours, as demonstrated in Zhang's Example 8) provides an improvement in the overall risks of adverse consequences of SE (such as brain damage, permanent disability, and death) at eight hours and beyond." Ex. 1041 ¶ 48.

However, Dr. Sankar responds that "Zhang does not state or provide any indication that multiple administration can prolong an effect from a single administration, let alone a prolonged effect on SE." Ex. 2014 ¶ 95 (citing Ex. 1009 ¶ 74). That is, Dr. Sankar is disagreeing with Dr. Rogawaski because Zhang lacks evidence of a prolonged effect, so the 5 hour improvement in Zhang need not necessarily extend to eight hours.

In this Ground, the sole reference relied upon by Petitioner for obviousness is Zhang, and we will therefore not import teachings from references relied upon in the other Grounds. We are persuaded by Dr. Sankar's statements that there would not have been a reasonable expectation of success from Zhang in improvement after 8 hours as Zhang shows diminishing results from ganaxolone after 5 hours and no evidence of efficacy beyond 5 hours. Ex. 2014 ¶¶ 94–97. Dr. Rogawski provides no specific evidence in Zhang supporting the position that the reduction of adverse consequences would reasonably be expected to occur.

Accordingly, we find that Petitioner does not show, by a preponderance of the evidence, that claim 22 of the ’817 patent is unpatentable as obvious over Zhang.

b. Claims 25–31

As we have already discussed, Zhang teaches “[s]eizure disorders that may be treated with the substituted β-cyclodextrin-ganaxolone injectable formulation include status epilepticus.” Ex. 1009 ¶ 70. Zhang teaches that a “substituted β-cyclodextrin-ganaxolone injectable formulation is administered as an intravenous infusion dose.” *Id.* ¶ 76. Zhang teaches a range of doses of about 1 mg/kg to about 20 mg/kg of ganaxolone. *See id.* ¶ 84; Ex. 1003 ¶ 103. Zhang expressly suggests treatment of human patients. Ex. 1009 ¶ 22.

We therefore agree with Petitioner that Zhang suggests intravenous treatment using a ganaxolone formulation in human patients with status epilepticus that overlap with the recitation of ganaxolone ranges recited in claims 25–31. *See, e.g.*, Ex. 1009 ¶¶ 22, 70, 76; Ex. 1003 ¶ 103.

We are not persuaded by Patent Owner’s assertion that Zhang lacks motivation for the specific dose ranges for treating SE recited in claims 25–31 because Zhang teaches a broader, overlapping range. *See* PO Resp. 57–58. “A *prima facie* case of obviousness typically exists when the ranges of a claimed composition overlap the ranges disclosed in the prior art.” *In re Peterson*, 315 F.3d 1325, 1329 (Fed. Cir. 2003). “The point of our overlapping range cases is that, in the absence of evidence indicating that there is something special or critical about the claimed range, an overlap suffices to show that the claimed range was disclosed in—and therefore obvious in light of—the prior art." *E.I. du Pont de Nemours & Co. v. Synvina C.V.*, 904 F.3d 996, 1008 (Fed. Cir. 2018).

That Zhang's overlapping range of effective amounts would suggest the particular values recited in claims 25–31 is supported by Dr. Rogawski, who states "assuming a single bolus dose per day as taught by Zhang, one could calculate the total dose in mg per day for an average human patient, weighing 70 kg," as shown in the table below.

Claim(s)	**Amount Administered Per Day (range in mg)**	**Amount Administered to a 70 kg Patient Per Day (range in mg/kg)**
24	150–1000	2.14–14.29
25	500–1000	7.14–14.29
26	625–650	8.93–9.29
27	650–675	9.29–9.64
28	675–700	9.64–10
29	625–700	8.93–10
30	700–725	10–10.36
31	1025–1050	14.64–15

Ex. 1003 ¶ 103. Dr. Rogawski states that "Zhang's bolus doses of 1 mg/kg to 20 mg/kg, for a 70 kg patient, encompass each of the amounts recited in claims 24–31." *Id.* ¶ 104.

We also find unpersuasive Patent Owner's argument that there would not have been a reasonable expectation of success in using the dosages recited in Zhang. *See* PO Resp. 58–59. As discussed above, we interpret the word "patient" to encompass a variety of patients including veterinary patients.

Zhang teaches that the "effective amount of ganaxolone will be selected by those skilled in the art depending on the particular patient and the disease." Ex. 1009 ¶ 24. Zhang teaches that "'an effective amount' . . . can vary from subject to subject, due to variation in metabolism of ganaxolone, age, weight, general condition of the subject, the condition being treated, the severity of the condition being treated, and the judgment of the prescribing physician." *Id*. Given this disclosure, Zhang recognizes that ganaxolone "may be effective at a different concentration in different formulations, but that is just a property of the particular known material, subject to conventional experimentation." *Almirall, LLC v. Amneal Pharm. LLC*, 28 F.4th 265, 273 (Fed. Cir. 2022).

Zhang demonstrates that ganaxolone was effective in treating mice with SE, teaching "[g]anaxolone at 12 and 15 mg/kg produced qualitatively similar results, with both dose levels producing a very strong reduction in SE amplitude overall." Ex. 1009 ¶ 134. Dr. Rogawski states that the

> experimental approach used is, in my opinion, generally considered to be predictive of clinical efficacy. I previously explained that the rat pilocarpine model is a well accepted model in the field . . . in my opinion, there is a reasonable expectation that a treatment which demonstrates improvement in animals experiencing SE as in the examples of Zhang . . . will also demonstrate some improvement in SE when administered to humans suffering from the same.

Ex. 1041 ¶ 56. Dr. Rogawski further states "dogs are susceptible to status epilepticus and when this occurs, they are in need of treatment. Many types of dogs have body weights that would be similar to the weight range of humans." *Id*. ¶ 30. Dr. Rogwaski states "[t]hat the specific doses set forth in claims 25–31 may be higher than the dose that would likely be given to a rat does not suggest to a person of skill in the art that the term 'patient' in the claims is limited to humans. Those doses could apply to any number of species." *Id*. In deposition, Dr. Rogawski stated "at the School of Veterinary Medicine at the University of Minnesota, we do treat animals with status epilepticus, and [the veterinarians] routinely describe their experiences." Ex. 2024, 27:10–13.

We recognize that Dr. Sankar states that Zhang's "result would not have provided a reasonable expectation of success for the specific doses of claims 25-31 in a human patient." Ex. 2014 ¶ 105. Dr. Sankar states "[t]he lack of efficacy data in humans is significant because of the difficulties in treating SE." *Id*. However, Dr. Rogawski states the "experimental approach used is, in my opinion, generally considered to be predictive of clinical efficacy." Ex. 1041 ¶ 56. Dr. Rogawski states "in my opinion, there is a reasonable expectation that a treatment which demonstrates improvement in animals experiencing SE as in the examples of Zhang . . . will also demonstrate some improvement in SE when administered to humans suffering from the same." *Id*.

Before determining whether claims 25–31 of the '817 patent would have been obvious over Zhang, we must consider the evidence above together with the objective indicia evidence, which we will analyze below.

c. *Objective Indicia analysis*

Objective indicia "must always when present be considered" in the overall obviousness analysis. *Bristol-Myers Squibb Co. v. Teva Pharms. USA, Inc.*, 752 F.3d 967, 977 (Fed. Cir. 2014) But "they do not necessarily control the obviousness determination." *Id*. at 977. Indeed, "a strong showing of obviousness may stand 'even in the face of considerable evidence'" of objective indicia. *ZUP, LLC v. Nash Mfg., Inc.*, 896 F.3d 1365, 1374 (Fed. Cir. 2018).

"For objective evidence of secondary considerations to be accorded substantial weight, its proponent must establish a nexus between the evidence and the merits of the claimed invention." *Wyers v. Master Lock Co.*, 616 F.3d 1231, 1246 (Fed. Cir. 2010) (brackets omitted). "Where the offered secondary consideration actually results from something other than what is both claimed and *novel* in the claim, there is no nexus to the merits of the claimed invention." *In re Huai-Hung Kao*, 639 F.3d 1057, 1068 (Fed. Cir. 2011).

(1) Unexpected results

Patent Owner asserts that the results reported in the Marinus Phase II Clinical Trial are unexpected and Petitioner does not dispute that the precise dosing regimen used in the clinical trial obtained unexpected results. PO Resp. 41–49; Sur-Reply 11; Reply 18, 22–24; *see, e.g.*, Ex. 2007, 31 ¶¶ 8–10.

Claim 22 of the '817 patent reasonably encompasses administration of any dose of ganaxolone using any regimen of intravenous administration to any patient. *See, e.g.*, Ex. 1042, 46:4–8. Claims 25–31 of the '817 patent reasonably encompass treatment of any patient using any regimen of intravenous administration as limited to the particular doses recited in the claims. For example, the broadest dose recited of 500 mg to 1000 mg/day ganaxolone is in claim 25 while claim 30 recites a narrower dose range of 700 mg to 725 mg/day ganaxolone.

We agree with Petitioner that claims 22 and 25–31 of the ’817 patent lack a nexus with the asserted unexpected results. This is evident from the Epilepsia[10] paper, the Rogawski Declaration, the Sankar Deposition and the Hulihan Declaration cited by Patent Owner. *See* PO Resp. 42.

The Epilepsia paper discussed the method of administration of ganaxolone, stating that “ganaxolone was studied in the following three dose cohorts: 500mg/day (low), 650mg/day (medium), and 713mg/day (high, the most ganaxolone that could be administered while staying within the daily Captisol® limit).” Ex. 2003, 2383. The Epilepsia paper states “[i]n all dose cohorts, ganaxolone infusion was initiated as an IV bolus (over 3 min) with continuous infusion of decreasing infusion rates for 48–96h followed by an 18-h taper.” *Id.* The Epilepsia paper explained that “[t]hese infusion parameters allowed for rapid loading of ganaxolone (plasma concentration ~900ng/ml) designed to abort SE, followed by maintenance doses aimed at sustaining seizure control.” *Id.*

The Epilepsia paper also reported results from the administration of ganaxolone in the SE patients, stating “[a]ll three dose cohorts received a similar initial ganaxolone bolus (25–30mg), resulting in rapid reductions in seizure burden within the first 15min of infusion.” Ex. 2003, 2386. The Epilepsia paper noted that

> Sustained clinical response was associated with maintaining ganaxolone plasma concentrations ≥500ng/ml. Ganaxolone plasma levels ≥500ng/ml were achieved only briefly during the initial bolus for patients in the medium-dose cohort, whereas those in low- and high-dose cohorts maintained such levels for

[10] Vaitkevicius et al., *Intravenous ganaxolone for the treatment of refractory status epilepticus: Results from an open-label, dose-finding, phase 2 trial*, 63 Epilepsia 2381–91 (2022); Ex. 2003.

> 4 and 8 h, respectively. The greatest decrease in seizure burden was seen when ganaxolone concentrations were maintained at ≥500ng/m for at least 8 h.

Id. The Epilepsia paper concluded that "[t]hese data support that the ganaxolone regimen administered in the high-dose cohort (713mg/day) may reduce or prevent escalation of care to third-line IV anesthetics." *Id*. at 2388.

Dr. Rogawski states regarding the Epilepsia paper that the "specific dosing regimens resulted in certain clinical results described in the publication. However, ganaxolone could be administered according to many other regimens that would result in the same or different total daily doses." Ex. 1041 ¶ 66. Dr. Rogawski notes that Patent Owner "provides no information as to whether the results described in the *Epilepsia* paper would be obtained if the bolus is eliminated." *Id*. Dr. Rogawski concludes that

> To the extent that any of these results would have been unexpected at all, they are tied to Marinus's [Petitioner's] specific dosing regimens, which Ovid [Patent Owner] does not claim. Ovid's claims are much broader and encompass countless possible dosing regimens. In short, Ovid's claims are not commensurate in scope with Marinus's clinical trial results.

Id. ¶ 70.

The Hulihan Declaration also explains the results from the trial reported in the Epilepsia paper and states

> we unexpectedly discovered that SE could be rapidly and persistently suppressed using less ganaxolone and for a much shorter period of time than was anticipated. We determined that administering ganaxolone as an intravenous bolus and continuous intravenous infusion to achieve and maintain a ganaxolone plasma concentration of about 500 ng/ml to about 1000 ng/ml for about 8 to about 12 hours resulted in rapid and persistent suppression of SE, without recurrence of SE when the rate of ganaxolone infusion was reduced and treatment was terminated.

Ex. 2007, 31 ¶ 8. That is, Dr. Hulihan found the particular administration regimen using a bolus and intravenous infusion resulting in a particular plasma concentration necessary to obtain the unexpected suppression of SE using ganaxolone. *See id*.

Dr. Hulihan explains that "targeting a transient ganaxolone plasma concentration of about 500 ng/ml to about 1000 ng/ml ***was not enough*** to result in persistent suppression of SE. We found that it is important to maintain that ganaxolone plasma concentration for about 8 to about 12 hours to achieve persistent suppression" Ex. 2007, 31 ¶ 9 (emphasis added). That is, Dr. Hulihan found it insufficient to simply administer ganaxolone to a particular plasma concentration but rather found that the particular dosing regimen mattered.

None of claims 22 and 25–31 of the '817 patent recite either a bolus or require particular plasma concentrations of ganaxolone described by Dr. Hulihan as necessary to obtain the unexpected suppression of SE. *See* Ex. 1041 ¶ 67. And while claims 22 and 25–31 do require administration of ganaxolone intravenously, the claims do not limit the administration to a rate that would maintain a ganaxolone plasma concentration within the 500 ng/ml to 1000 ng/ml for 8 to 12 hours, as described to be necessary to achieve the unexpected SE suppression in the Hulihan Declaration. *See* Ex. 2007, 31 ¶ 8; Ex. 1041 ¶ 67.

The understanding that claims 22 and 25–31 of the '817 patent lacks any particular dosage regimen is supported by Patent Owner's expert, Dr. Sankar, who in response to the question "[b]ut in that context then there's no dosage regimen described in claim 22 or incorporated into claim 22; correct?" answered "yeah." Ex. 1042, 46:4–8. Dr. Sankar also agreed that each of claims 25–31 provide ranges but not any more detail. *Id*. at 46:10–49:2.

Dr. Sankar also recognized the impact of a bolus dose, noting

> Q. What do you understand when it talks about the "rapid loading of ganaxolone"?
> A. It's to achieving a therapeutic level promptly.
> Q. And why is that important?
> A. Because status is an emergency. You want to gain control of it.
> Q. And is that why it's important to use the three-minute bolus?
> A. Yes. You want to achieve the blood level quickly and then you try to maintain the drip rate.

Ex. 1042, 53:24–54:9. When asked if a spike in blood levels to 900 nanograms per ml was due to bolus administration of ganaxolone, Dr. Sankar stated regarding getting the spike that "You won't without a bolus." *Id*. at 55:1–10. When asked "what do you understand is occurring then at times zero or shortly thereafter?" Dr. Sankar stated "[s]hortly thereafter the seizure burden comes down dramatically as quantified by EEG." *Id.* at 56:9–12. In response to the follow-up asking "is that because the bolus had been delivered?", Dr. Sankar stated "[c]orrect." *Id.* at 56:13–15. When asked specifically "if you limited the bolus would you think that the patients would achieve that end point?" Dr. Sankar stated "My suspicion is they would not have been controlled as rapidly." *Id.* at 67:22–68:2. And Dr. Sankar answered "Yes" to the question "[s]o changing the dosage regimen could change the results." *Id*. at 68:10–12.

We need not rely upon the Baughman Declaration[11] for our analysis because we find the evidence already discussed sufficient to show that there is no nexus between claims 22 and 25–31 of the '817 patent and the results in the Epilepsia paper. *See* Ex. 1058, Ex. 2025.

We are not persuaded by Patent Owner that claims and the specification in the '817 patent that are ***entirely silent*** regarding elements such as how the drug is administered (*see, e.g.*, Sur-Reply 13–14) may be presumed to have a nexus with unexpected results, as here, tied to specific administration profiles with specific amounts that result in specific blood levels shown to be significant in the treatment of SE with ganaxolone by intravenous administration after bolus. *See* Ex. 2007, 31 ¶ 8; Ex. 1041 ¶ 67. *See In re GPAC Inc.*, 57 F.3d 1573, 1580 (Fed. Cir. 1995) ("For objective [evidence of secondary considerations] to be accorded substantial weight, its proponent must establish a nexus between the evidence and the merits of the claimed invention.")

A presumption of nexus is only appropriate if the patentee shows that the asserted objective evidence is tied to a specific product and that product embodies the claimed features, and is coextensive with them. *Fox Factory, Inc. v. SRAM, LLC*, 944 F.3d 1366, 1373 (Fed. Cir. 2019). The coextensiveness requirement is not met simply by showing that "the patent claims broadly cover the product that is the subject of the evidence of secondary considerations." *Id.* at 1377. Here the evidence including Dr.

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[11] Patent Owner moves to exclude the Baughman Declaration. Paper 34, 8–13. That motion is dismissed as to the Baughman Declaration, which we do not rely on herein, as explained in our concurrently filed order.

Sankar's statements demonstrates that Patent Owner fails to show results that are coextensive with claims 22 and 25–31 of the '817 patent.

Thus, a preponderance of the evidence of record does not support a finding of nexus between the unexpected results in the Epilepsia paper and claims 22 and 25–31 of the '817 patent.

(2) Long felt need

To establish a long-felt need, three elements must be proven: First, the need must have been a persistent one that was recognized by ordinarily skilled artisans. *In re Gershon*, 372 F.2d 535, 538 (CCPA 1967). Second, the long-felt need must not have been satisfied by another before the invention. *See Newell Companies, Inc. v. Kenney Mfg. Co.*, 864 F.2d 757, 768 (Fed. Cir. 1988) ("[O]nce another supplied the key element, there was no long-felt need or, indeed, a problem to be solved . . . ."). Third, the invention must, in fact, satisfy the long-felt need. *In re Cavanagh*, 436 F.2d 491, 496 (CCPA 1971).

Patent Owner asserts the "prior art is replete with statements recognizing the long-felt need for a drug that could treat SE. Reddy acknowledged that one of the 'key areas of unmet clinical needs in epilepsy' included 'new AEDs [antiepileptic drug] for drug-resistant seizures and SE.' Ex. 1004, 18306." PO Resp. 50. Patent Owner asserts "Ganaxolone administered according to the methods in claims 22 and 25–31 filled this unmet need. Ex. 2014, ¶¶ 119-121." *Id*. Patent Owner asserts that "[a]s shown in Petitioner's phase 2 clinical trial, the claimed improvement and dose ranges of ganaxolone resulted in rapid seizure cessation and reduction in seizure burden. Ex. 2003, 2386, 2388." *Id*. Patent Owner asserts "[n]either Zhang nor Saporito provides any information about using ganaxolone to treat SE in humans and therefore could not supply 'the key element' to meet this long-felt need." Sur-Reply 18.

Petitioner agrees that "there is no question that SE is a 'difficult to treat seizure disorder with limited treatment options.'" Reply 30 (citing Ex. 1011, 1). However, Petitioner asserts that they, not Patent Owner "worked for years to provide that treatment option, as evidenced by the cited art. '[O]nce another supplied the key element, there [is] no long-felt need.'" *Id*. (citing *Newell*, 865 F.2d at 768) (alterations in original).

As discussed already, Zhang teaches "[s]eizure disorders that may be treated with the substituted β-cyclodextrin-ganaxolone injectable formulation include status epilepticus." Ex. 1009 ¶ 70. Zhang teaches that a "substituted β-cyclodextrin-ganaxolone injectable formulation is administered as an intravenous infusion dose." *Id*. ¶ 76. Zhang teaches a range of doses of about 1 mg/kg to about 20 mg/kg of ganaxolone. *See id*. ¶ 84; Ex. 1003 ¶ 103. Zhang expressly suggests treatment of human patients. Ex. 1009 ¶ 22. Additionally, the Marinus Orphan Drug Press Release states

> the U.S. Food and Drug Administration (FDA) has granted Orphan Drug Designation to the intravenous (IV) formulation of its CNS-selective GABAA modulator, ganaxolone, for the treatment of status epilepticus. A phase 1 clinical trial evaluating the safety, tolerability and pharmacokinetics of ganaxolone IV is expected to initiate in the first half of 2016.

Ex. 1011, 1.

While both parties acknowledge there is evidence supporting a long felt need existed, we agree with Petitioner that the prior art of Zhang and Marinus Orphan Drug Press Release reasonably supply the key element satisfying the need before the invention. Patent Owner is incorrect in asserting Zhang fails to suggest human treatment of SE as Zhang is clearly focused on such human treatment. Zhang's background is entirely focused on SE patients and the failure of then-conventional compounds, stating "[t]here are currently no good treatment options for these patients" where the patients discussed are clearly human patients. Ex. 1009 ¶ 1. Zhang expressly states that there was a "need for additional treatments for seizure disorders such as status epilepticus . . . This disclosure fulfills this need and provides additional advantages." *Id.* ¶ 4. Zhang identifies humans as patients expressly. *Id.* ¶ 22.

That Zhang's examples were performed using an animal model of SE in order to obtain a patent does not detract from the clear teachings in Zhang to use ganaxolone to treat human patients with SE. Ex. 1009 ¶¶ 1, 4, 22, 70. And the Marinus Orphan Drug Press Release confirms treatment of humans with SE with ganaxolone prior to Patent Owner's invention. Ex. 1011, 1. Dr. Rogawski further states "[b]y the time that the contents of the '817 patent specification first published, Marinus had already begun its clinical trials of an intravenous formulation of ganaxolone with the intention of developing it for the treatment of SE. This research was publicly disclosed in Tsai, (Ex. 1020)." Ex. 1041 ¶ 13. "It is well established that a patent may be secured, and typically is secured, before the conclusion of clinical trials." *In re Montgomery*, 677 F.3d at 1382. *Montgomery* also approvingly cites to the Manual of Patent Examining Procedure § 2107.03 (8th ed., rev.6, Sept. 2007) ("[I]f an applicant has initiated human clinical trials for a therapeutic product or process, [Patent & Trademark] Office personnel should presume that the applicant has established that the subject matter of that trial is reasonably predictive of having the asserted therapeutic utility."). *Id.* at 1382–83.

Thus, a preponderance of the evidence of record does not support a finding of an unmet long-felt need for the invention recited in claims 22 and 25–31 of the ’817 patent.

(3) Failure by others

Patent Owner asserts compounds “such as lorazepam and diazepam, are among the ‘first-line therapies’ for treating SE, but as Petitioner admits, ‘[a] significant drawback to these therapies is that their efficacy decreases dramatically as the duration of SE increases, and there can be a complete loss of efficacy after a certain amount of time.’” PO Resp. 51 (citing Pet. 11) (alteration in original). Patent Owner asserts that “allopregnanolone was seen as a promising candidate for treating SE in humans, however, it did not receive FDA approval because it failed to meet its primary efficacy endpoints in a Phase 3 clinical trial. Ex. 2015, 23:6-24:1.” *Id*.

Petitioner asserts “‘rather than show failure by others, the evidence of record suggests success by others.’ And any supposed failure by non-ganaxolone drugs is irrelevant.” Reply 30.

We agree with Petitioner as there is no evidence that Zhang, the prior art reference at issue, failed. We agree with Petitioner that failure of other unrelated drugs has minimal, if any, relevance to the question of failure by others when Zhang demonstrates success. Zhang demonstrates efficacy in an animal model as discussed above. Zhang teaches that when tested in animals “[g]anaxolone at 12 and 15 mg/kg produced qualitatively similar results, with both dose levels producing a very strong reduction in SE amplitude overall.” Ex. 1009 ¶ 134.

There also is not evidence that the clinical trial in Marinus Orphan Drug Press Release failed. In deposition, Dr. Sankar acknowledged regarding this document that "it seems logical that it was a promising approach to that, otherwise they wouldn't have gone into phase 1." Ex. 1042, 87:2–4. Dr. Rogawski noted that

> Dr. Sankar believed the work described in the '817 patent "probably inspired Marinus to take it and start doing some clinical experimentation," and was not aware of any earlier publications describing it (Sankar Deposition Transcript, Ex. 1042, page 24, lines 12–18 and page 25, lines 21–22). This cannot be the case, however, as, again, this work was first published *after* Marinus had begun its clinical trials.

Ex. 1041 ¶ 14.

Thus, rather than show failure by others, a preponderance of the evidence of record supports a conclusion of success by others prior to the invention of claims 22 and 25–31 of the '817 patent.

d. Conclusion

We balance the objective indicia with Petitioner's obviousness position to determine whether a preponderance of the evidence shows whether claims 25–31 of the '817 patent would have been obvious over Zhang.

We find that the direct suggestion of Zhang to treat human and veterinary patients with overlapping ranges of amounts of ganaxolone intravenously for SE provides reason and reasonable expectation of success in doing so. When we balance Zhang's disclosures with evidence of unexpected results that lack nexus to the recitations of claims 25–31 of the '817 patent, and unpersuasive evidence of long felt need and absence of failure by others, we determine that a preponderance of the evidence supports finding claims 25–31 of the '817 patent obvious.

Accordingly, we find that Petitioner has demonstrated, with a preponderance of the evidence, that claims 25–31 of the ’817 patent would have been unpatentable as obvious over Zhang. As discussed above, Petitioner has not demonstrated that claim 22 of the ’817 patent would have been unpatentable as obvious over Zhang alone.

E. Ground 3 – Obviousness over Zhang, Saporito P1, Saporito P2

1. Petitioner’s positions

a. Petitioner’s position on claim 22

Petitioner asserts “Zhang is described above.” Pet. 50. Petitioner asserts that Saporito P1 teaches

> rats were administered lithium chloride (127 mg/kg) and pilocarpine (50 mg/kg) to induce convulsive SE (CSE), a type of SE in which the seizures experienced by the subject are convulsions. Rats were then administered ganaxolone (formulated at a concentration of 2.5 mg/mL in 30% CAPTISOL®) at a dose of 6, 9, or 12 mg/kg at the time of seizure onset (time zero) and 15, 30, and 60 minutes after seizure onset.

Id. (citing Ex. 1013; Ex. 1003 ¶ 112). Petitioner asserts “IV administered ganaxolone ‘halted CSE and produced a dose-dependent reduction in seizures associated with CSE when administered at 3 different doses over four separate time points.’” *Id.* at 50–51 (citing Ex. 1013; Ex. 1003 ¶ 113). Petitioner asserts that Saporito P2 performs a similar experiment on rats to induce SE and showed “the administration of ganaxolone treated SE, by terminating or stopping SE.” *Id.* at 53 (citing Ex. 1014, Ex. 1003 ¶ 118).

Petitioner specifically points out, pertinent to claim 22, that the Saporito P1 “poster also shows that the IV administered ganaxolone promotes survival, measured 24 hours after the onset of CSE.” Pet. 51, 56 (citing Ex. 1013, Fig. 3).

b. Petitioner's position on claims 25–31

Petitioner asserts a "POSA would have combined Zhang's teachings (e.g., as described in Section VIII.A.8) with Saporito P1 and/or P2's teachings, as described above. And, for at least the reasons described in Section VIII.B, it would have been with a reasonable expectation of success. Therefore, claims [25]–31 would have been obvious." Pet. 57 (footnote omitted).

2. Patent Owner's positions

a. Patent Owner's position on claim 22

Patent Owner asserts that "Petitioner has failed to establish that Saporito P1 and Saporito P2 were sufficiently publicly accessible before the priority date of the '817 patent. Saporito P1 and Saporito P2 are not dated, and there is no indication on their face that they were presented at the 2016 AAN meeting." PO Resp. 62 (citing Ex. 2016, 48:8–22, 71:16–23; Ex. 1013; Ex. 1014). Patent Owner asserts that the "Saporito Declaration does not indicate how many people attended the poster sessions other than the sessions being 'well attended' (Ex. 1015, ¶¶ 10, 21) or that the declarant 'interacted with a number of meeting attendees' at Poster Session IV (*id.*, ¶ 15)." *Id.* Patent Owner notes that when asked in November 2023 in deposition how many people had attended a poster session in April 2016, Dr. Saporito "admitted that he could not remember or provide a more specific number of attendees." *Id.* (citing Ex. 2016, 64:23-65:20).

Patent Owner also asserts "[t]he Petition and the Saporito Declaration also provide no explanation of whether Exhibits 1013 and 1014 are the same posters that were presented at the April 2016 AAN meeting apart from merely stating that they are." PO Resp. 63. Patent Owner asserts "[n]either the Saporito declaration nor the Supplemental Saporito declaration provides any chain of custody for the files that Petitioner alleges are the posters that were presented or indicates whether the declarant possessed these files or if someone else did." *Id*. Patent Owner asserts that because "Saporito P1 contains a different set of authors than the meeting brochure that purportedly discusses the underlying poster begs the question—unanswered by Petitioner—as to whether Saporito P1 is the poster that was presented at the 2016 AAN meeting." *Id*. at 64. Patent Owner similarly asserts "[t]hat Saporito P2 lists none of the authors listed in the AAN Meeting Brochure again begs the question as to whether Saporito P2 is the poster that was presented at the 2016 AAN meeting." *Id*. at 65.

Patent Owner asserts that "Zhang does not describe intravenously administering to a human patient suffering from SE a pharmaceutical composition comprising ganaxolone or a pharmaceutically acceptable salt thereof." PO Resp. 65 (citing Ex. 2014 ¶¶ 125–128). Patent Owner also asserts that "Saporito P1 and Saporito P2 also do not disclose intravenously administering to a human patient a pharmaceutical composition comprising ganaxolone or a pharmaceutically acceptable salt thereof." *Id*. at 65–66 (citing Ex. 2014 ¶ 126).

Patent Owner asserts "it is common knowledge to a POSA that animal models do not translate directly to human treatment." PO Resp. 66 (citing Ex. 2014 ¶ 127). Patent Owner asserts "doses of ganaxolone administered to rats cannot be directly compared with doses administered to humans because rats metabolize compounds at a much faster rate than humans by the hepatic microsomal system." *Id*. at 67.

Patent Owner also asserts "there is no motivation to combine different aspects of Zhang to arrive at the limitations recited in claim 22 (i.e., 'improvement to the patient after administration for more than 8 hours')." PO Resp. 67–68 (citing Ex. 2014, ¶¶ 123-24, 129). Patent Owner asserts neither Saporito reference "provides the additional details necessary to arrive at the claimed limitations, as both references provide only data from rats, which a POSA would have understood does not translate directly to human treatment. *Id*. (citing Ex. 2014, ¶ 130). Patent Owner also asserts "Zhang does not disclose "improvement to the patient after administration for more than 8 hours," as claim 22 requires. *Id*. (citing Ex. 2014, ¶ 132). Patent Owner asserts

> Petitioner's argument that "Saporito P1 shows that ganaxolone promoted survival in the rats, measured 24 hours after the onset of CSE, suggesting improvement after 8 hours" has no support in Dr. Rogawski's declaration or in Saporito P1 itself. Pet., 56; Ex. 2014, ¶ 136. Dr. Rogawski does not opine, and Saporito P1 does not disclose, that increased survival measured 24 hours after the onset of CSE represents an improvement in treating SE after 8 hours.

Id. at 70. Patent Owner asserts "Saporito P1's disclosures about 24-hour survival rate show no statistical significance of the data, unlike the data disclosed in Saporito P2." *Id*.

b. Patent Owner's position on claims 25–31

Patent Owner asserts "a POSA would not have been motivated to choose the specific dose ranges recited in claims 25–31 from the broader dose range disclosed in Zhang (1 mg/kg to 20 mg/kg)." PO Resp. 71 (citing Ex. 2014, ¶¶ 137–38). Patent Owner asserts that neither "Saporito P1 or Saporito P2 would cure the deficiency in Zhang by providing a motivation for choosing the dose ranges recited in claims 25–31." *Id*. Patent Owner asserts

> Petitioner has not shown that a POSA would have had a reasonable expectation of success in treating a human patient suffering from SE with the doses recited in claims 25–31 based on Zhang's disclosure. Nor would a POSA have had a reasonable expectation of success in doing so with the addition of Saporito P1 or Saporito P2. Ex. 2014, ¶¶ 139–40. Petitioner does not contend that Saporito P1 or Saporito P2 cure this deficiency in Zhang. Pet., 57.

Id. at 72.

3. *Analysis*

a. *Saporito Posters as prior art*

We first address the issue of whether a preponderance of the evidence demonstrates that the Saporito posters are prior art and printed publications. *Medtronic* explained that in *Klopfenstein*, a poster of slides was displayed at a professional conference but not distributed to the public and stated

> Under such a scenario, we identified the relevant factors to include: (1) "the length of time the display was exhibited," (2) "the expertise of the target audience" (to determine how easily those who viewed the material could retain the information), (3) "the existence (or lack thereof) of reasonable expectations that the material displayed would not be copied," and (4) "the simplicity or ease with which the material displayed could have been copied.". . . After reviewing these factors, we determined that the reference was sufficiently publicly accessible to count as a "printed publication" for the purposes of § 102(b).

Medtronic, Inc. v. Barry, 891 F.3d 1368, 1381–82 (Fed. Cir. 2018) (citing *In re Klopfenstein*, 380 F.3d 1345, 1347–50 (Fed. Cir. 2004)).

The AAN 2016 meeting program listed the Saporito posters as available with "Presenters Stand by Posters" and identified the posters as

P2.020 and P4.212. Ex. 1012, 48, 99. In the AAN 2016 meeting program and in Exhibit 1013, Saporito Poster 1 is identically titled "Ganaxolone Administered Intravenously Prevents Behavioral Seizures and Promotes Survival in the Rat Lithium-Pilocarpine Model of Status Epilepticus." In the AAN 2016 meeting program and in Exhibit 1014, Saporito Poster 2 is identically titled "Intravenous Administration of Ganaxolone Attenuates Electroencephalographic Seizures in a Diazepam Resistant Model of Status Epilepticus." Dr. Saporito states that "[t]he poster shown at page 1 of Ex. 1013 is the same document that I remember displaying as a poster, standing by, and discussing with attendees at the meeting" and makes a similar statement regarding Exhibit 2014. *See* Ex. 1033 ¶¶ 4, 7.

In deposition in response to a question regarding a difference in authorship between the abstract in the meeting program and the authorship recited on the Saporito Poster 1, which included Donna Henry as an author, Dr. Saporito answered "we probably decided that Donna should be one of the authors because of her contribution and her relationship with some of these scientists. So we probably added her at a later stage." Ex. 2016, 54:16–20. Dr. Saporito explained the reason that an author addition to the poster might not appear in the brochure is because the "brochure is derived from the abstract, which is submitted well, well before the meeting." *Id*. at 54:23–25.

Dr. Saporito states regarding the Saporito Poster 1 that (1) the "Poster Session II was held from 8:30am to 5:50pm on the 17th, with presenters, including myself, standing by the posters from 4pm to 5:30pm to interact with the Annual Meeting attendees who chose to stop and view the poster." Dr. Saporito similarly explained that Saporito Poster 2 was also presented during a poster session at the meeting. Ex. 1015 ¶¶ 9, 13. As to the audience expertise, Dr. Saporito stated (2) the AAN Annual Meeting "is attended by both clinicians practicing in neurology and researchers interested in looking for potential new treatments for neurological disorders. I would consider it to be an important and influential gathering in the neurology field." *Id.* ¶ 5. Dr. Saporito stated (3) "[a]s with Saporito Poster 1, because Saporito Poster 2 was shown and presented at the meeting, I understood that the information on the poster would no longer be considered confidential." *Id.* ¶ 13. Dr. Saporito stated (4) "[j]ust as they did for Saporito Poster 1 from Poster Session II, I recall that attendees took notes and photographs while looking at Saporito Poster 2." *Id.* ¶ 15.

Dr. Saporito's statements are further supported by Dr. Rogawski, who stated

> The 2016 annual meeting where the Saporito posters were presented was held April 15–21, 2016, at the Vancouver Convention Center in Vancouver, B.C. and had 11,670 attendees who were able to study the 2,700 poster presentations available for public viewing at the meeting. (2016 AAN Annual Report, Ex. 1019, page 12.) The Saporito posters were among those on public display and freely available to all 11,670 attendees.

Ex. 1003 ¶ 37. In deposition, Dr. Rogawski explained that the Declaration statement "was based on my personal knowledge and also reviewing information about the meeting . . . it was indicated on the website and . . . I believe I found an annual report from the society." Ex. 2015, 107:19–108:2.

We find Patent Owner's arguments unpersuasive, as at best, Patent Owner criticizes some aspects of the Declaration of Dr. Saporito such as the exact number of attendees, that Dr. Saporito only stated[12] the current Exhibits were the same as those presented without any chain of custody, and that there were some alleged discrepancies regarding authorship. PO Resp. 61–64. Patent Owner provides no evidence substantively rebutting any of Dr. Saporito's testimony addressing these issues (Ex. 1015,1033), but rather simply relies on attorney argument. "Attorney argument is not evidence." *Icon Health & Fitness, Inc. v. Strava, Inc.*, 849 F.3d 1034, 1043 (Fed. Cir. 2017).

We find that a preponderance of the evidence supports a determination that the Saporito posters were publicly available as they were presented at the AAN annual meeting, that these were exhibited to an audience comprising interested practitioners and researchers for a sufficient time to assimilate or copy the information, and that the poster presenters expected copying of their posters, and that actual photographs of the presentation occurred. Ex. 1015 ¶¶ 5–18.

b. Claim 22

We find that a preponderance of the evidence supports a determination that claim 22 would have been obvious over Zhang, Saporito P1, and Saporito P2.

As discussed above, Zhang suggests intravenous treatment using a ganaxolone formulation in human patients with status epilepticus. *See, e.g.*, Ex. 1009 ¶¶ 22, 70, 76.

[12] We note that Dr. Saporito stated the posters were the same "under the penalty of perjury." Ex. 1015, ¶ 2 , p. 9.

As to the limitation in claim 22 that ganaxolone administration results in improvement in the patient for more than 8 hours, Saporito P1 stated "[t]he number of rats that survived the treatment paradigm in each group was also recorded and compared between study groups and time points." Ex. 1013, 1. Saporito P1 stated "GNX IV promoted survival at rates similar to those observed with administration of allopregnanolone as measured at 24 hours after CSE onset in CSE induced rats." *Id*. Figure 3 of Saporito P1 is reproduced below:

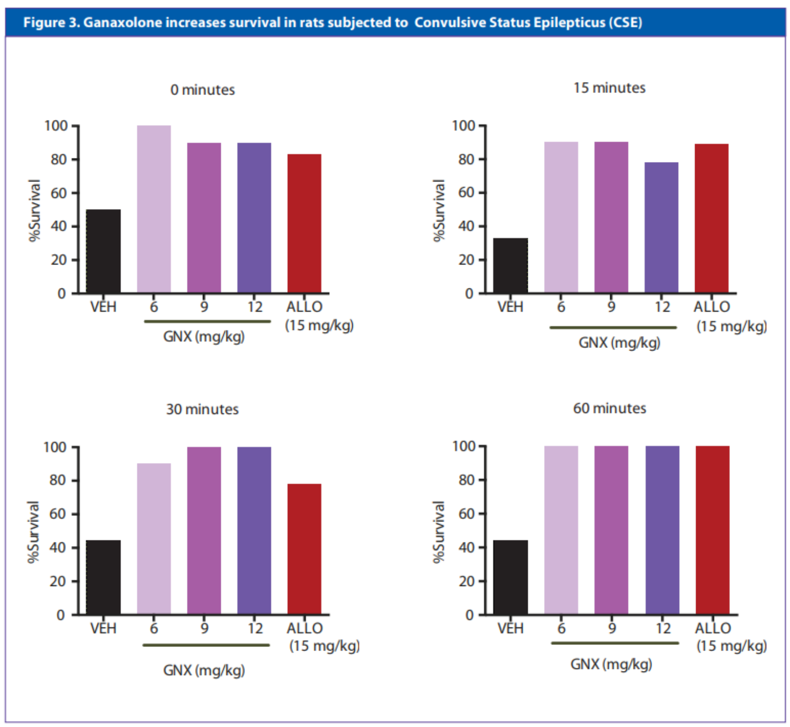

Figure 3 shows "Vehicle, ganaxolone or allopregnanolone were adminstered [sic] by IV bolus injection at time of CSE (time 0) and 15, 30, or 60 minutes after CSE. The study was terminated 24 hours after CSE onset and the number of surviving animals were quantitated." *Id*. Saporito P1 teaches the number of animals surviving after 24 hours increased after three different ganaxolone doses at four different timepoints of administration relative to untreated animals, providing evidence of improvement in the condition of the animals for more than 8 hours. *Id*. at Fig. 3.

In deposition, when asked "what is the other column, the 24-hour survival referring to?" Dr. Sankar stated "How many of them. As I indicated earlier a lot of animals die after this pilocarpine status. In my lab we often lost up to 40 percent of them to mortality so this is just survival." Ex. 1042, 32:13–18. Dr. Sankar also stated that "ganaxolone in the right dose can substantially salvage these animals." *Id*. at 33:4–5. When asked "what do you mean by salvage?" Dr. Sankar stated "[t]hat they survive overnight which may partly be because we were able to gain control of their seizures." *Id*. at 33:6–9. Dr. Sankar stated "in a study pertaining to treatment, if you increase survival that will be a beneficial outcome in treatment." *Id*. at 37:19–21.

Dr. Rogawski similarly testified that:

> Improvement beyond 8 hours is shown explicitly, for example, in the experimental results presented in Saporito Poster 1 . . . rats receiving a single bolus IV administration of ganaxolone at all doses (including the 12 mg/kg dose) administered at all time points after SE onset (including 15 and 60 minutes after SE onset) exhibited markedly increased rates of survival at 24-hours compared to vehicle, such that nearly all animals survived. (Ex. 1013, Methods, Figure 3, Conclusions).

Ex. 1041 ¶ 49. In deposition, when asked regarding Figure 3 of the Saporito Poster 1 "whether or not ganaxolone was blocking seizures; is that right?" Dr. Rogawski stated "one thing that's very, very clear here is that there was dramatic improvement in the animals that were treated with ganaxolone at all doses, at the 24-hour time point that was assessed here. Because less of them were damaged, it's clearly an improvement." Ex. 2015, 119:16–23; *cf. id*. at 121:24–122:4.

We are not persuaded by Patent Owner's argument that the ordinary artisan would have lacked motivation to treat SE with ganaxolone based on the disclosures of Zhang, Saporito P1, and Saporito P2. As already discussed, we do not agree with Patent Owner's interpretation of patient as limited to humans, but rather interpret the term "patient" consistent with the '817 patent to include "humans, canines, porcine, ungulates, rodents, poultry, and avian." Ex. 1001, 32:65–67. Zhang teaches to treat both humans and rodents with ganaxolone intravenously to mitigate SE. Ex. 1009 ¶¶ 22, 70, 76, 135.

And a preponderance of the evidence of record expressly teaches that ganaxolone as administered intravenously by Saporito P1 to rats with SE provided improvement to those rats for 24 hours, a period that is longer than 8 hours and thus "more than 8 hours" as recited in claim 22. Ex. 1013, Fig. 3; Ex. 1042, 37:19–21; Ex. 2015, 119:16–23.

We find that these disclosures would have provided motivation to treat non-human animals such as rats with SE with ganaxolone due to the beneficial effects. However, even if we limited the term "patients" to humans, we find that a preponderance of the evidence would also have provided motivation to treat humans with SE, as Zhang expressly suggests that patients may be humans. Ex. 1009 ¶ 22 ("A 'patient' is a human."). Dr. Rogawski explained that

> those of skill in the art understood on the priority date of the '817 patent, and still understand, the rat lithium-pilocarpine model to be a widely accepted model to identify potential treatments for SE. Halting behavioral SE in the model indicates that a test substance is effective in treating the SE condition.

Ex. 1003 ¶ 115. This supports our conclusion that the ordinary artisan interested in treating SE would have been motivated by successful results in animal models to apply the same effective intravenous ganaxolone treatment to human populations with SE as "there is a reasonable expectation that a treatment which demonstrates improvement in animals experiencing SE as in the examples of Zhang, as well as Saporito P1 and P2, will also demonstrate some improvement in SE when administered to humans suffering from the same." Ex. 1041 ¶ 56; *Acorda Therapeutics, Inc. v. Roxane Labs., Inc.*, 903 F.3d 1310, 1333–34 (Fed. Cir. 2018) ("This Court has long rejected a requirement of '[c]onclusive proof of efficacy' for obviousness." (alteration in original)).

We are also unpersuaded by Patent Owner's assertion regarding an absence of a reasonable expectation of success. *See, e.g.*, Sur-Reply 22. We find that a preponderance of the evidence demonstrates efficacy of ganaxolone in the rat pilocarpine model system. Ex. 1013, Fig. 3; Ex. 1042, 37:19–21; Ex. 2015, 119:16–23. This evidence provides a reasonable expectation of success for use of the treatment in other patients including humans for the same periods of time of treatment. Saporito P1 provides specific evidence showing that ganaxolone administration improved patient outcomes for periods of 8 or more hours. Ex. 1013, Fig. 3; Ex. 1014, Fig. 4. "[O]bviousness does not require absolute predictability of success . . . all that is required is a reasonable expectation of success." *In re Kubin*, 561 F.3d 1351, 1360 (Fed. Cir. 2009) (emphasis omitted, second alteration in original).

c. Claims 25–31

As we have already discussed above, Zhang teaches a range of doses of about 1 mg/kg to about 20 mg/kg of ganaxolone for treatment of humans with diseases including status epilepticus by intravenous infusion. *See*

Ex. 1009 ¶¶ 22, 70, 76, 84; Ex. 1003 ¶ 103. As also discussed above, Zhang teaches the "effective amount of ganaxolone will be selected by those skilled in the art depending on the particular patient and the disease." Ex. 1009 ¶ 24. Zhang demonstrates that ganaxolone was effective in treating mice with SE just as Saporito P1 and Saporito P2 demonstrate ganaxolone was effective in treating mice with SE. *See id.* ¶ 134; Ex. 1013, Figs. 2, 3; Ex. 1014, Figs. 3, 4. We also note that the remainder of our discussion of Zhang with regard to these claims above remains applicable to this ground, with the additional support of efficacy shown in the Saporito P1 and Saporito P2 posters.

As already discussed, we recognize that Dr. Sankar states that Zhang's "result would not have provided a reasonable expectation of success for the specific doses of claims 25-31 in a human patient." Ex. 2014 ¶ 105. Dr. Sankar states "[t]he lack of efficacy data in humans is significant because of the difficulties in treating SE." *Id.* However, Dr. Rogawski states the "experimental approach used is, in my opinion, generally considered to be predictive of clinical efficacy." Ex. 1041 ¶ 56. Dr. Rogawski states "in my opinion, there is a reasonable expectation that a treatment which demonstrates improvement in animals experiencing SE as in the examples of Zhang . . . will also demonstrate some improvement in SE when administered to humans suffering from the same." *Id.*

Combined with the disclosures of Zhang, Saporito P1, and Saporito P2, we find Dr. Rogawski more persuasive and credible regarding the obviousness of using the dosages of claims 25–31 of the '817 patent for treatment of SE.

d. Objective Indicia analysis

We incorporate our objective indicia analysis as discussed above in V.D.3.c. for claims 22 and 25–31 of the ’817 patent.

e. Conclusion

We balance the objective indicia with Petitioner’s obviousness position to determine whether a preponderance of the evidence shows whether claims 22 and 25–31 of the ’817 patent would have been obvious over Zhang, Saporito P1, and Saporito P2. We find that the direct suggestion of Zhang to treat human and veterinary patients with overlapping ranges of amounts of ganaxolone intravenously for SE provides reason and reasonable expectation of success in doing so. We find that the disclosure in Saporito P1 showing efficacy of ganaxolone in survival of rats with SE for 24 hours provides a reasonable expectation of success in for the recitation in claim 22 requiring improvement for more than 8 hours and Saporito P2 shows efficacy of ganaxolone in SE as well.

As we balance the disclosures of Zhang, Saporito P1, and Saporito P2 with evidence of unexpected results that lack nexus to the recitations of claims 22 and 25–31 of the ’817 patent, and unpersuasive evidence of long felt need and absence of failure by others, we determine that a preponderance of the evidence supports finding claims 22 and 25–31 of the ’817 patent obvious.

Accordingly, we find that Petitioner has demonstrated, with a preponderance of the evidence, that claims 22 and 25–31 of the ’817 patent would have been unpatentable as obvious over Zhang, Saporito P1, and Saporito P2.

F. Ground 4 – Obviousness over Zhang, Marinus Orphan Drug Press Release, Marinus Phase I Press Release

1. Petitioner's positions

a. Petitioner's position on Claim 22

Petitioner asserts "Zhang . . . is described above." Pet. 57. Petitioner asserts that Marinus Orphan Drug Press Release teaches

> "the U.S. Food and Drug Administration (FDA) granted Orphan Drug Designation to the intravenous (IV) formulation of its CNS-selective $GABA_A$ modulator, ganaxolone, for the treatment of [SE]." It also discloses that "[a] Phase 1 clinical trial evaluating the safety, tolerability and pharmacokinetics of ganaxolone IV is expected to initiate in the first half of 2016.

Id. at 58 (citing Ex. 1011; Ex. 1003 ¶ 125) (footnote omitted, alterations in original). Petitioner asserts Marinus Phase I Press Release teaches the company had "dosed the first subject in its Phase 1 clinical trial of ganaxolone IV, an intravenous (IV) formulation' for the treatment of SE." *Id.* (citing Ex. 1022; Ex. 1003 ¶ 126). Petitioner asserts that Marinus Phase I Press Release "also discloses that the study would 'include a dose escalation of a bolus dosage of ganaxolone IV and a bolus dose of ganaxolone IV, followed by a continuous infusion.'" *Id.* (footnote omitted). Petitioner asserts that for "at least the same reasons set forth for claim 1 . . . claim 22 would have been obvious over Zhang and Marinus's Orphan Drug Press Release and/or Marinus's Phase I Press Release." *Id.* at 63–64 (footnote omitted).

b. Petitioner's position on claims 25–31

Petitioner asserts

> [f]or at least the same reasons set forth for claim 1, a POSA would have combined Zhang's teachings (e.g., as described in Section VIII.A.8) with the teachings of Marinus's

> Orphan Drug and Phase I Press Releases, as described above. And, for at least the reasons described in Section VIII.B, it would have been with a reasonable expectation of success. Therefore, claims 24–31 would have been obvious over Zhang and Marinus's Orphan Drug Press Release and/or Marinus's Phase I Press Release.

Pet. 64–65.

2. Patent Owner's positions

a. Patent Owner's position on claim 22 and 25–31

Patent Owner asserts a "POSA would not have been motivated to combine Zhang with Marinus's Orphan Drug Press Release and/or Marinus's Phase I Press Releases to arrive at the limitations recited in claims 22 and 25-31. Ex. 2014, ¶¶ 142-161." PO Resp. 72. Patent Owner asserts

> nothing in Zhang's disclosure would have motivated a POSA to meet the limitation of claim 22, as Zhang does not disclose "improvement to the patient after administration for more than 8 hours." Ex. 2014, ¶ 145. Nor would Zhang have motivated a POSA to choose the specific dose ranges disclosed in claims 25-31. Ex. 2014, ¶ 154. And a POSA would not have been motivated by Zhang to seek out additional publications related to the treatment of SE by administering ganaxolone to a patient in need because Zhang's disclosure is limited to data in rats. Ex. 2014, ¶¶ 146, 155.

Id. at 73. Patent Owner asserts

> Petitioner fails to show that a POSA would have had a reasonable expectation of success that intravenous administration of ganaxolone would result in "improvement to the patient after administration for more than 8 hours" or that administration of ganaxolone at the doses recited in claims 25-31 would treat SE in human patients based on Zhang in combination with Marinus's Orphan Drug Press Release and/or Marinus's Phase I Press Release. Ex. 2014, ¶¶ 147, 156.

Id. at 74. Patent Owner asserts "the fact that the FDA granted orphan drug status to Marinus for the use of ganaxolone to treat SE, and the fact that Marinus was allowed to proceed to Phase I clinical trials for this use, say nothing about the efficacy of the drug. Ex. 2014, ¶¶ 150-51, 159-60." *Id*. at 75.

3. Analysis

a. Claim 22

As we discussed above with regard to Zhang, a preponderance of the evidence does not show that Zhang would inherently, necessarily, or obviously result in 8 hour improvement in any particular species of patient after administration of ganaxolone. As we have noted, that conclusion is supported by Petitioner's own expert, Dr. Rogawski, who stated with regard to the claimed 8 hour improvement that "I think it's possible that there could be improvement, there might not be improvement. When one is treating a medical condition, there's always a degree of variability and one doesn't expect a certain outcome." Ex. 2015, 155:9–16. Dr. Rogawski therefore acknowledges that improvement at the 8 hour timepoint is not necessarily the natural result of ganaxolone treatment and may not necessarily occur.

We have considered both Marinus Orphan Drug Press Release and Marinus Phase I Press Release. Both references further support Zhang's suggestion to treat humans with SE, and Marinus Phase I Press Release emphasizes the importance of bolus dosages in these studies. *See* Ex. 1011, Ex. 1022.

However, neither Marinus Orphan Drug Press Release nor Marinus Phase I Press Release suggests any particular degree of treatment success and neither of these disclosures provides any discussion of the length of time of improvement that would be expected to result from treatment of SE with ganaxolone. *See* Ex. 1011, Ex. 1022.

As with the obviousness ground over Zhang alone, discussed earlier, we are persuaded by Dr. Sankar's statements that there would not have been a reasonable expectation of success from Zhang in improvement after 8 hours as Zhang shows diminishing results from ganaxolone after 5 hours and no evidence of efficacy beyond 5 hours. Ex. 2014 ¶¶ 94–97. Dr. Rogawski provides no specific evidence in Zhang supporting the position that the reduction of adverse consequences would reasonably be expected to occur. Nor is there evidence from Marinus Orphan Drug Press Release nor Marinus Phase I Press Release addressing this limitation of claim 22.

Accordingly, we find that Petitioner does not show, by a preponderance of the evidence, that claim 22 of the '817 patent is unpatentable as obvious over Zhang, Marinus Orphan Drug Press Release and Marinus Phase I Press Release.

b. Claims 25–31

As we have already discussed above, Zhang teaches a range of doses of about 1 mg/kg to about 20 mg/kg of ganaxolone for treatment of humans with diseases including status epilepticus by intravenous infusion. *See* Ex. 1009 ¶¶ 22, 70, 76, 84; Ex. 1003 ¶ 103. As also discussed above, Zhang teaches the "effective amount of ganaxolone will be selected by those skilled in the art depending on the particular patient and the disease." Ex. 1009 ¶ 24. Zhang demonstrates that ganaxolone was effective in treating mice with SE. *See* Ex. 1009 ¶ 134. Marinus Orphan Drug Press Release teaches "it has dosed the first [human] subject in its phase 1 clinical trial of ganaxolone IV, an intravenous (IV) formulation of Marinus' CNS-selective GABAA modulator, for the treatment of status epilepticus (SE)." Ex. 1022, 1.

Thus, Zhang, Marinus Orphan Drug Press Release and Marinus Phase I Press Release all reasonably suggest treatment of SE with ganaxolone in dose amounts effective for human patients. Ex. 1009 ¶ 24; Ex. 1022, 1; Ex. 1011, 1.

As already discussed, we recognize that Dr. Sankar states that Zhang's "result would not have provided a reasonable expectation of success for the specific doses of claims 25-31 in a human patient." Ex. 2014 ¶ 105. Dr. Sankar states "[t]he lack of efficacy data in humans is significant because of the difficulties in treating SE." *Id*. However, Dr. Rogawski states the "experimental approach used is, in my opinion, generally considered to be predictive of clinical efficacy." Ex. 1041 ¶ 56. Dr. Rogawski states "in my opinion, there is a reasonable expectation that a treatment which demonstrates improvement in animals experiencing SE as in the examples of Zhang . . . will also demonstrate some improvement in SE when administered to humans suffering from the same." *Id*.

Combined with the disclosures of Zhang, Marinus Orphan Drug Press Release and Marinus Phase I Press Release, we find Dr. Rogawski more persuasive and credible regarding the obviousness of using the dosages of claims 25–31 of the '817 patent for treatment of SE in human patients.

c. Objective Indicia analysis

We incorporate our objective indicia analysis as discussed above in V.D.3.c. for claims 22 and 25–31 of the '817 patent.

d. Conclusion

We balance the objective indicia with Petitioner's obviousness position to determine whether a preponderance of the evidence shows whether claims 25–31 of the '817 patent would have been obvious over Zhang, Marinus Orphan Drug Press Release, and Marinus Phase I Press Release. We find that the direct suggestion of Zhang to treat human and veterinary patients with overlapping ranges of amounts of ganaxolone intravenously for SE provides reason and reasonable expectation of success in doing so. Zhang also directly suggests optimizing this dose for patients. This reasoning is strongly bolstered by the teaching of Marinus Orphan Drug Press Release and Marinus Phase I Press Release to treat humans with SE.

As we balance the disclosures of Zhang, Marinus Orphan Drug Press Release, and Marinus Phase I Press Release with evidence of unexpected results that lack nexus to the recitations of claims 25–31 of the '817 patent, and unpersuasive evidence of long felt need and absence of failure by others, we determine that a preponderance of the evidence supports finding claims 25–31 of the '817 patent obvious.

Accordingly, we find that Petitioner has demonstrated, with a preponderance of the evidence, that claims 25–31 of the '817 patent would have been unpatentable as obvious over Zhang, Marinus Orphan Drug Press Release, and Marinus Phase I Press Release. As discussed above, Petitioner has not demonstrated that claim 22 of the '817 patent would have been unpatentable as obvious over Zhang, Marinus Orphan Drug Press Release, and Marinus Phase I Press Release.

G. Ground 7 – Enablement

1. Petitioner position

Petitioner asserts

> the ’817 patent specification contains very little data to support its claims related to treating SE with ganaxolone. If Ovid [Patent Owner] takes the position that the references applied to Challenged Claims 1–3, 10–17, and 19–31, above—which contain as much and, in fact, *more* data to support treating SE with ganaxolone—do not anticipate and/or render them obvious by arguing that the references are not enabling, then the claims of the ’817 cannot be enabled.

Pet. 74. Petitioner asserts that “[c]laims 21–31 recite a particular dosage form (intravenous), and in some cases, particular concentrations, dosing amounts, or patient outcomes.” *Id.* at 77. Petitioner asserts the “’817 patent (unlike Marinus’s publications), does not contain any working examples of treating SE (in a rat model or otherwise) using an intravenous dose form.” *Id.* at 77–78. Petitioner asserts, regarding doses and patient outcomes, that “a person of skill in the art would certainly not have arrived at them based on the only dose (20 mg/kg) for which data is presented the ’817 patent (Example 3) that could possibly suggest efficacy for treating SE in a rat.” *Id.* at 78.

Petitioner asserts that Patent Owner’s “expert, who did not opine on enablement, believes that ‘absent human data the patent cannot claim treating a human patient.’” Reply 31 (citing Ex. 1042, 41:19–21, 78:14–19) (footnote omitted). Petitioner asserts “[i]f the prior art data in rats is not sufficient to render the claims obvious, then the data in the specification itself is also not sufficient to enable them.” *Id.*

2. Patent Owner's position

Patent Owner asserts

> Instead of providing a cogent analysis grounded in the factors from *In re Wands*, 858 F.2d 731 (Fed. Cir. 1988), Petitioner makes the conclusory assertion that "a person of skill in the art would certainly not have arrived at [the claimed concentrations, dosing amounts and/or patient outcomes] based on the only dose (20 mg/kg) for which data is presented [in] the '817 patent (Example 3) that could possibly suggest efficacy for treating SE in a rat." Pet., 78 (citing Ex. 1003, ¶¶ 158-160). And Dr. Rogawski's declaration fares no better as it merely repeats verbatim the attorney argument in the Petition. *Compare* Pet., 78 *with* Ex. 1003, ¶ 159.

PO Resp. 76. Patent Owner asserts

> Petitioner has failed to put forth any evidence that treating SE with ganaxolone was unpredictable, that a large quantity of experimentation would have been necessary to show that ganaxolone was effective to treat SE, that the scope of claims 22 and 25-31 are unduly broad, that the guidance in the '817 patent's specification is insufficient, or that the prior art identifies particular concerns with the claimed treatment method.

Id. at 77. Patent Owner acknowledges "[a]ll Petitioner points to is a lack of working examples. See Petition, 78." *Id*.

Patent Owner asserts that the only evidence Petitioner "offers is its contention that Dr. Sankar 'believes that "absent human data the patent cannot claim treating a human patient."' Reply, 31. But Dr. Sankar offered no opinions on enablement in his declaration." Sur-Reply 25.

3. Analysis

The enablement requirement asks whether "the specification teach[es] those in the art to make and use the invention without undue experimentation." *In re Wands*, 858 F.2d 731, 737 (Fed. Cir. 1988). "[T]he specification must enable the full scope of the invention as defined by its claims. The more one claims, the more one must enable." *Amgen Inc. v. Sanofi*, 598 U.S. 594, 610 (2023).

"To prove that a claim is invalid for lack of enablement, a challenger must show . . . that a person of ordinary skill in the art would not be able to practice the claimed invention without 'undue experimentation.'" *Alcon Research Ltd. v. Barr Labs., Inc.*, 745 F.3d 1180, 1188 (Fed. Cir. 2014). In analyzing undue experimentation, we consider:

> (1) the quantity of experimentation necessary, (2) the amount of direction or guidance presented, (3) the presence or absence of working examples, (4) the nature of the invention, (5) the state of the prior art, (6) the relative skill of those in the art, (7) the predictability or unpredictability of the art, and (8) the breadth of the claims.

In re Wands, 858 F.2d at 737. There is no necessary requirement that mandates a *Wands* analysis for an enablement analysis as shown by *Amgen*, which affirmed a non-enablement determination without explicitly addressing any *Wands* factors.

We do not find that a preponderance of the evidence supports non-enablement of claims 22 and 25–31 of the '817 patent.

a. Wands factors based on cited evidence

(1) Breadth of claims

Claims 22 and 25–31 are narrowly drawn to a particular drug compound, ganaxolone, and a particular disorder, status epilepticus. *See, e.g.*, Ex. 1001, 40:8–31.

(2) Skill in the Art

The level of skill in the art has already been addressed above in Section IV and was determined to be very high.

(3) Working Examples

Dr. Rogawski states that the '817 patent "includes an experimental Example that utilizes the rat lithium-pilocarpine model to investigate the effect of administering ganaxolone. In this case, intraperitoneally." *See* Ex. 1003 ¶ 155; Ex. 1001, 35:30–37:15. Example 3 shows "Dose-response evaluation of treatment during benzodiazepine-resistant status epilepticus suggests dose-dependent efficacy of ganaxolone (i.p.)—significant ($p<0.02$) improvement in protection at 20 mg/kg." Ex. 1001, 37:8–11.

(4) The amount of direction or guidance presented

The '817 patent provides disclosure of methods "for treating an epileptic disorder by administering ganaxolone to a patient in need thereof." Ex. 1001, 6:27–29. The '817 patent discusses doses and modes of treatment with ganaxolone including doses overlapping the currently claimed values. *Id*. at 8:11–44. The '817 patent discusses time frames for improvement after treatment including the 8 hour timepoint. *Id*. at 24:45–49.

(5) State of the prior art and the unpredictability in the art

Petitioner provides no evidence that the treatment of status epilepticus patients with ganaxolone was unpredictable. *See generally* Pet. In contrast, Dr. Rogawski states the prior art "demonstrates, based on extensive experimentation in the rat lithium-pilocarpine model, that ganaxolone can treat SE. And, in particular, that ganaxolone administered intravenously was effective for treating SE." Ex. 1003 ¶ 154. Dr. Rogawski further states "based on the results seen in the rat lithium-pilocarpine model, a person of skill in the art would reasonably have expected ganaxolone to be effective for treating SE in humans." *Id*.

(6) Quantity of Experimentation

There is limited discussion of the quantity of experimentation in the record, but the working example in the ’817 patent and in Zhang, Saporito P1, and Saporito P2 reasonably suggest that the quantity of experimentation in animals is reasonable. In contrast, Marinus Orphan Drug Press Release and Marinus Phase I Press Release suggest a high quantity of experimentation for treatment in humans as clinical trials were necessary to evaluate ganaxolone in humans. *See* Ex. 1011, 1022.

(7) Analysis

Considering the *Wands* factors as a whole, we find a narrow breadth of the claims, a reasonable amount of direction and guidance provided by the ’817 patent, an absence of unpredictability and prior art evidence showing efficacy, a working example, and a high level of skill in the art, balanced against a high quantity of experimentation. Based on our consideration of the entirety of the evidence, we find that a preponderance of the evidence supports the conclusion that undue experimentation would not have been required to make and use the invention commensurate in scope with the claims of the ’817 patent.

VI. CONCLUSION

After considering the evidence and arguments before us in the complete trial record, we conclude that Petitioner has demonstrated, by a preponderance of the evidence, that Challenged Claims 22 and 25–31 of the ’817 patent are unpatentable.[13]

[13] Should Patent Owner wish to pursue amendment of the Challenged Claims in a reissue or reexamination proceeding subsequent to the issuance

In summary:

Claim(s)	**35 U.S.C. §**	**Reference(s)/Basis**	**Claim(s) Shown Unpatentable**	**Claim(s) Not shown Unpatentable**
22, 25–31	102	Zhang		22, 25–31
22, 25–31	103	Zhang	25–31	22
22, 25–31	103	Zhang, Saporito P1, Saporito P2	22, 25–31	
22, 25–31	103	Zhang, Marinus Orphan Drug Press Release, Marinus Phase I Press Release	25–31	22
22, 25–31	112(a)	Enablement		22, 25–31
Overall Outcome			22, 25–31	

of this decision, we draw Patent Owner's attention to the April 2019 Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding, 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. §§ 42.8(a)(3), (b)(2).

VII. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 22 and 25–31 of the ’817 patent are determined to be unpatentable; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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